

BFS Traversal

Step 1: Start at E

Legend

Complete

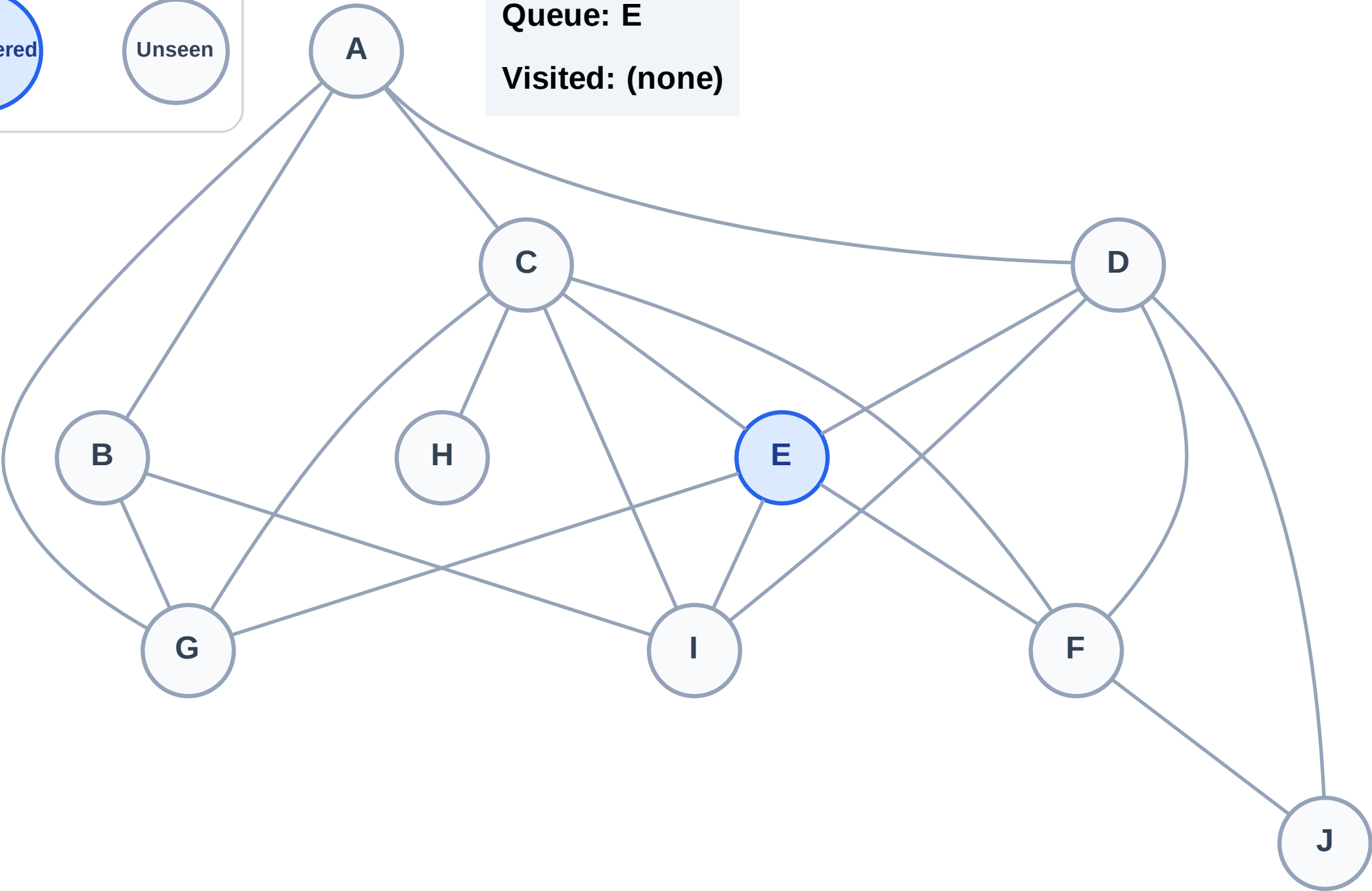
Processing

Discovered

Unseen

Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

Complete

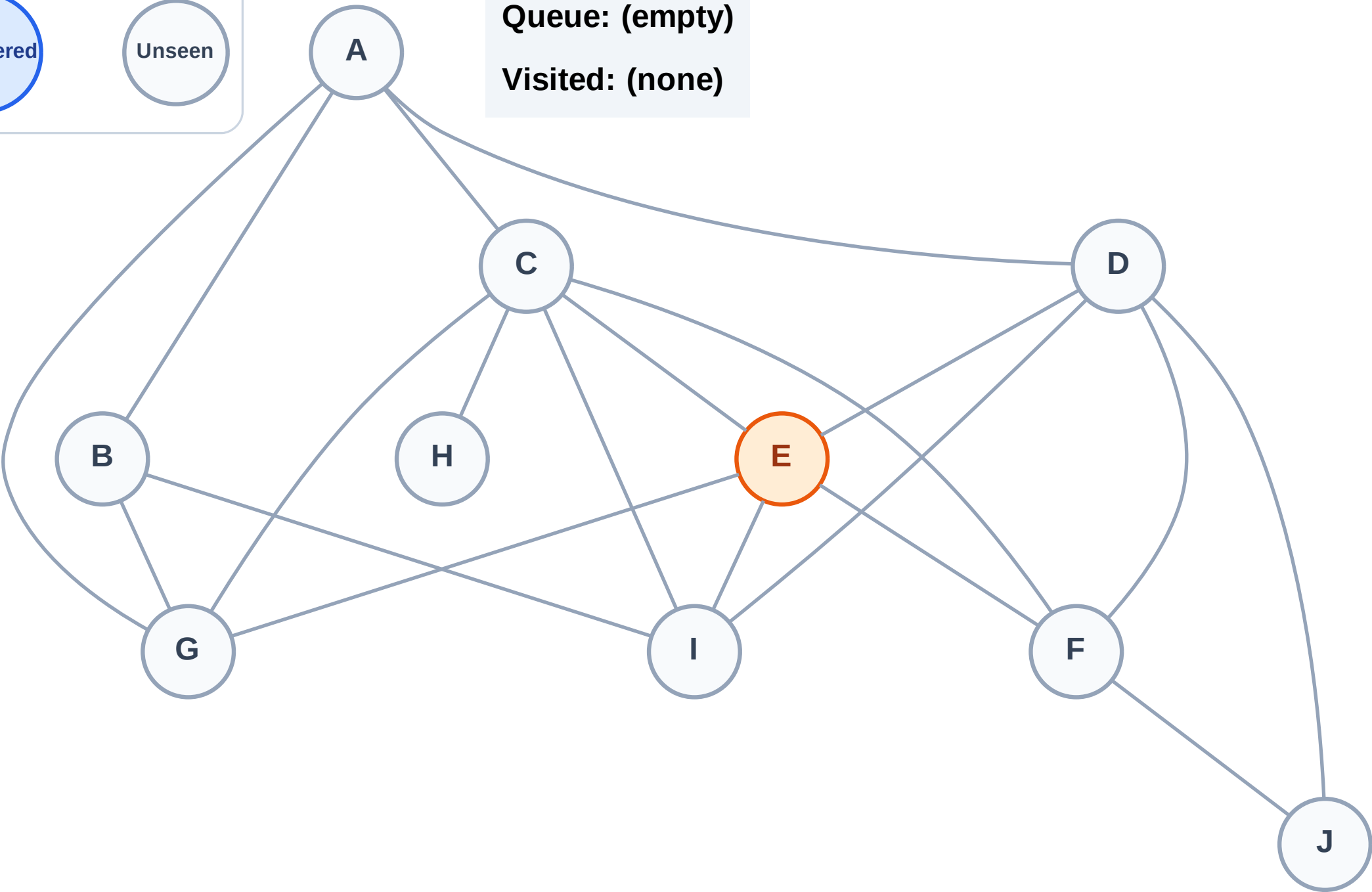
Processing

Discovered

Unseen

Queue: (empty)

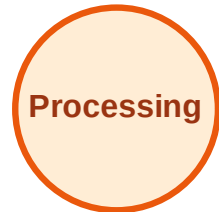
Visited: (none)



BFS Traversal

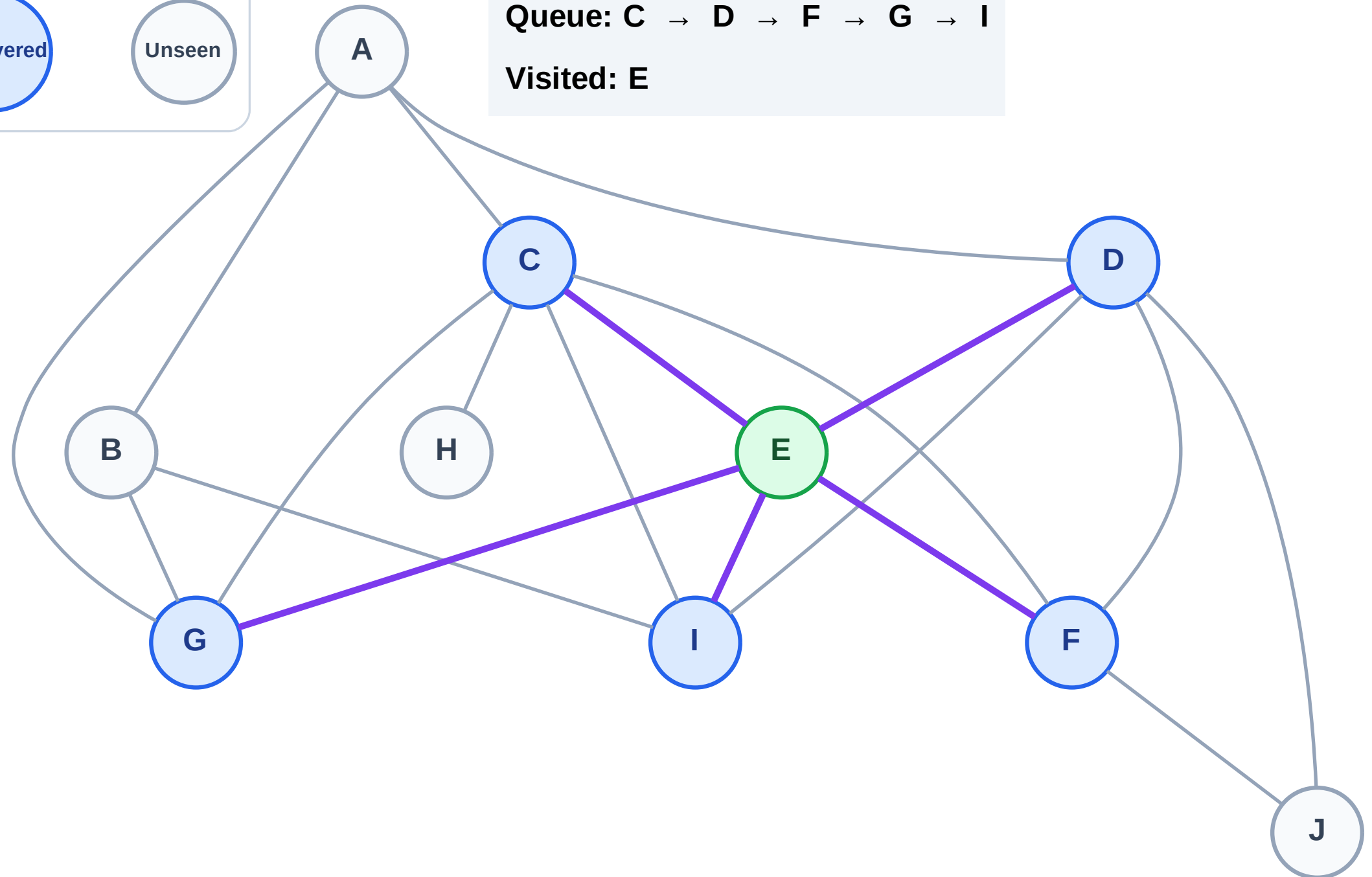
Step 3: Discover C, D, F, G, I from E

Legend



Queue: C → D → F → G → I

Visited: E



BFS Traversal

Step 4: Process node C

Legend

Complete

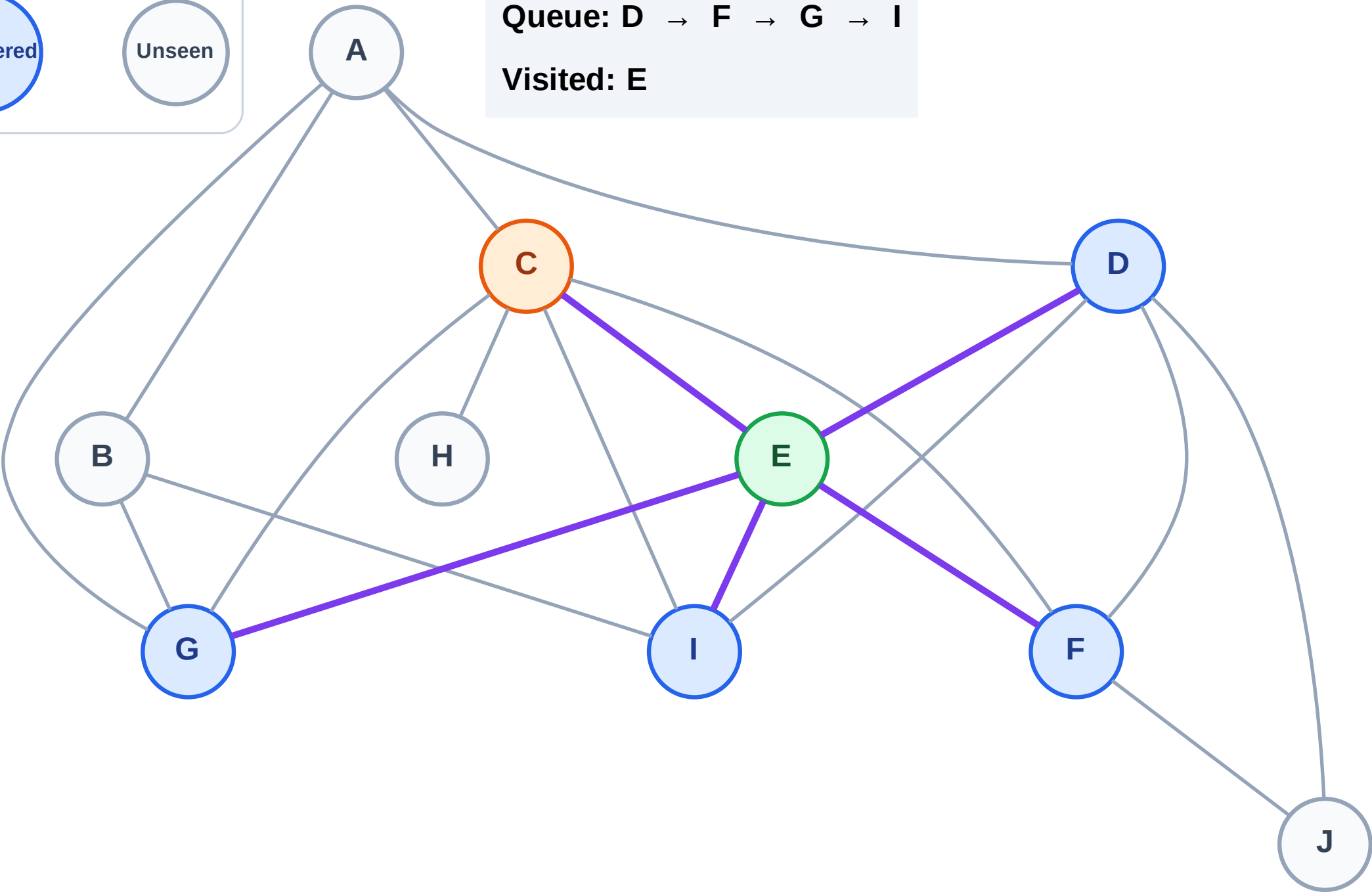
Processing

Discovered

Unseen

Queue: D → F → G → I

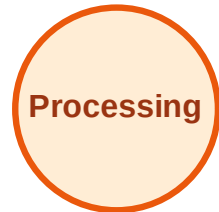
Visited: E



BFS Traversal

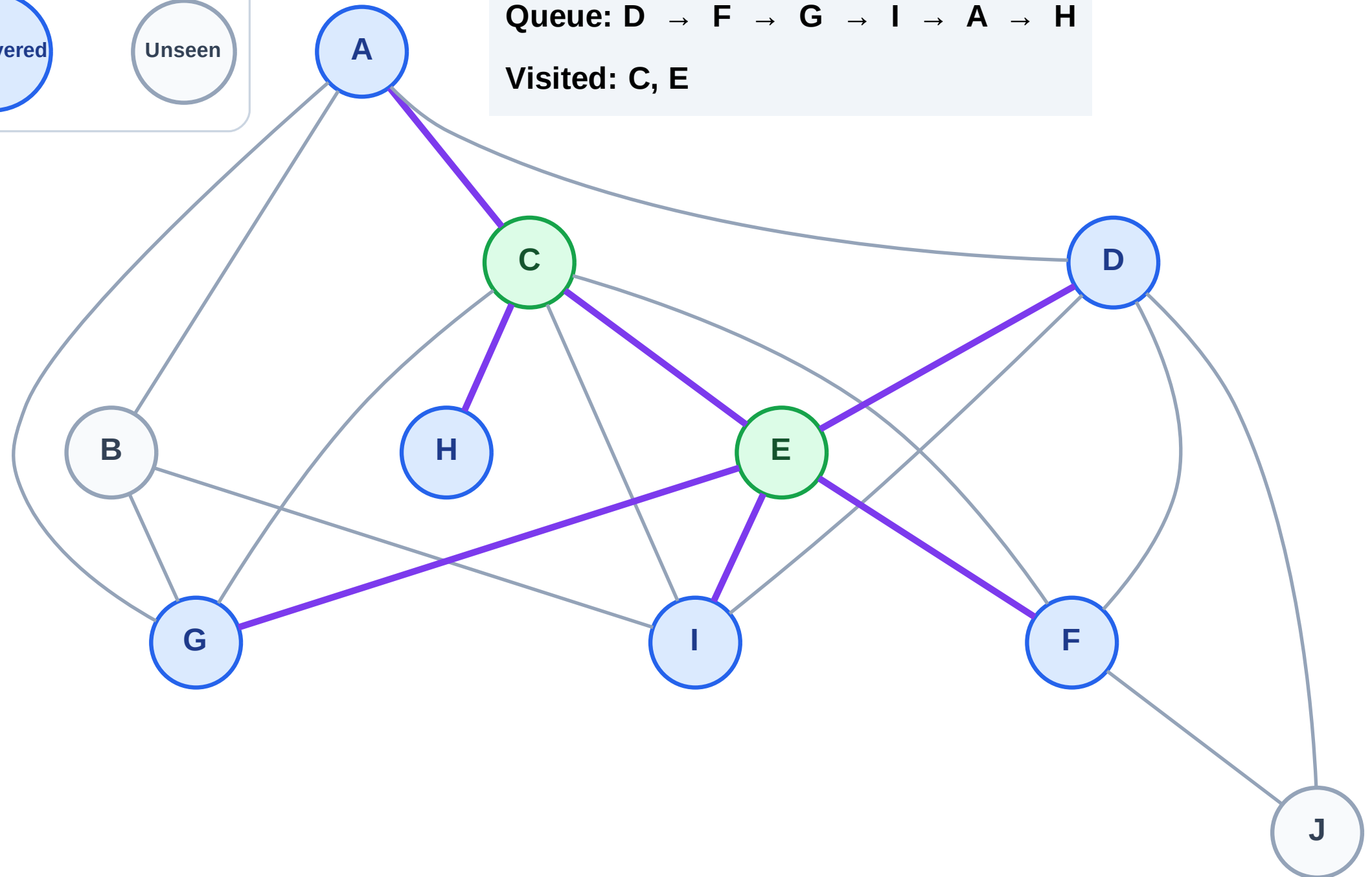
Step 5: Discover A, H from C

Legend



Queue: D → F → G → I → A → H

Visited: C, E



BFS Traversal

Step 6: Process node D

Legend

Complete

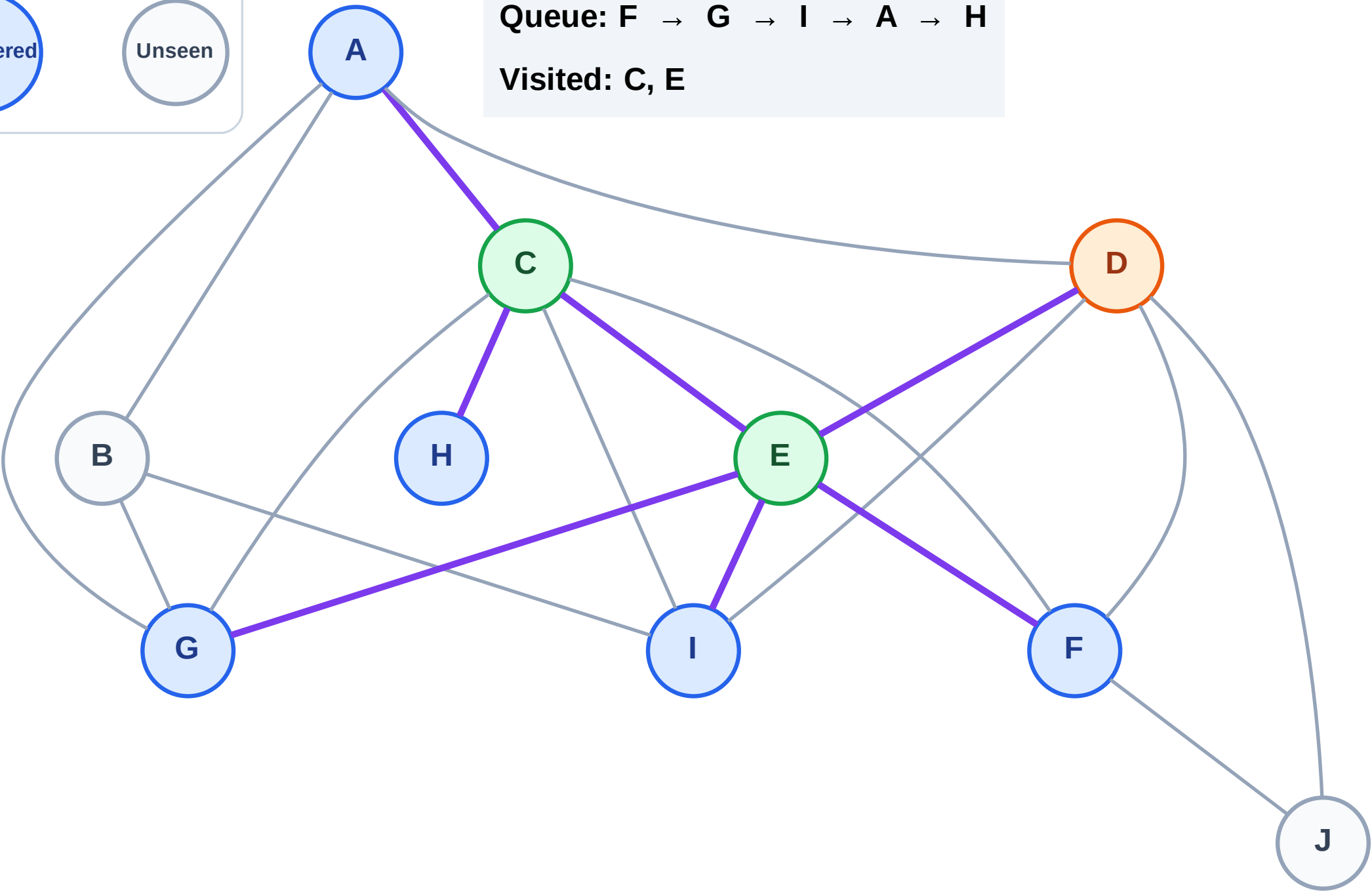
Processing

Discovered

Unseen

Queue: F → G → I → A → H

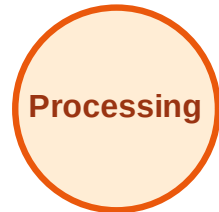
Visited: C, E



BFS Traversal

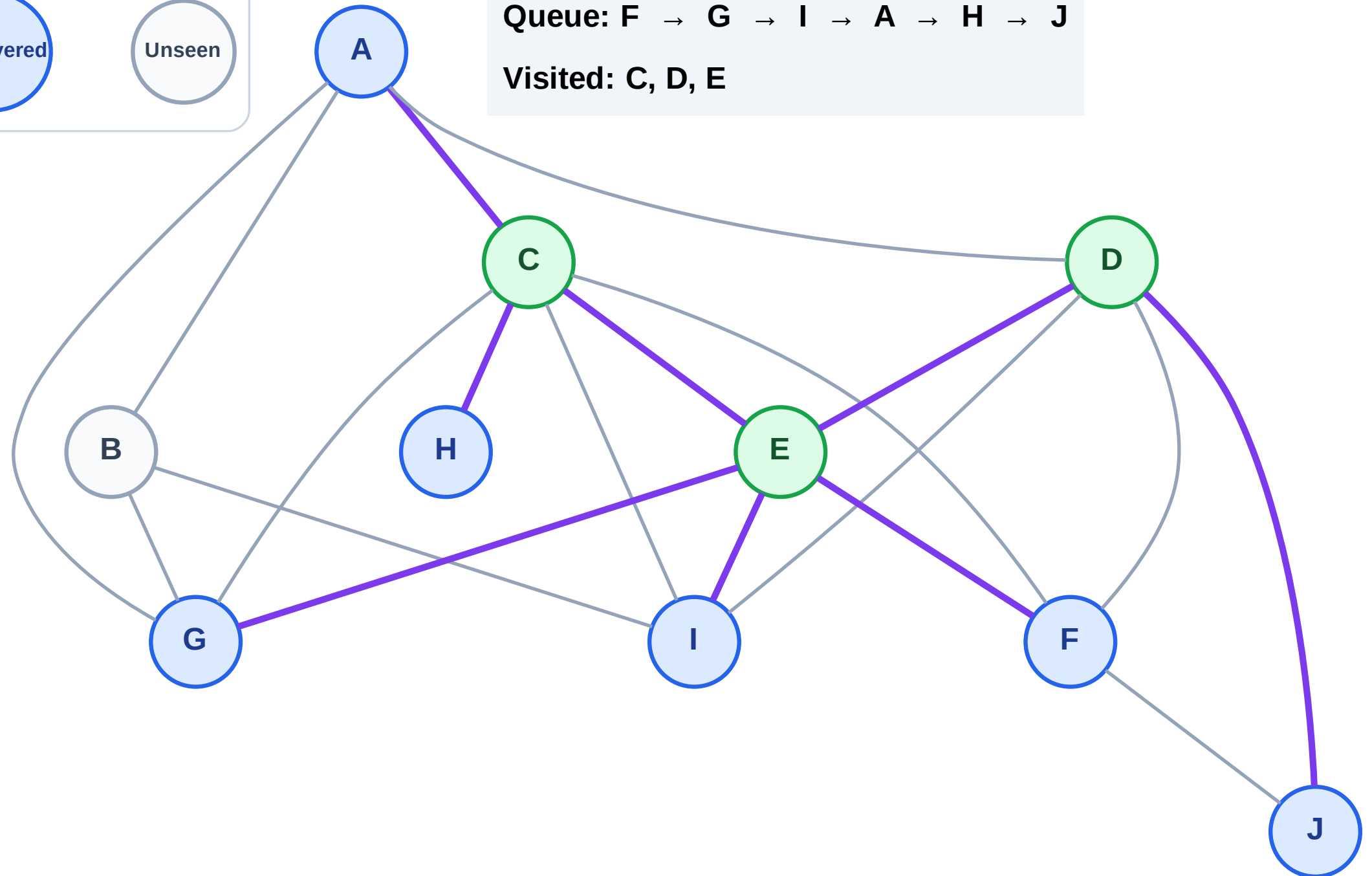
Step 7: Discover J from D

Legend



Queue: F → G → I → A → H → J

Visited: C, D, E



BFS Traversal

Step 8: Process node F

Legend

Complete

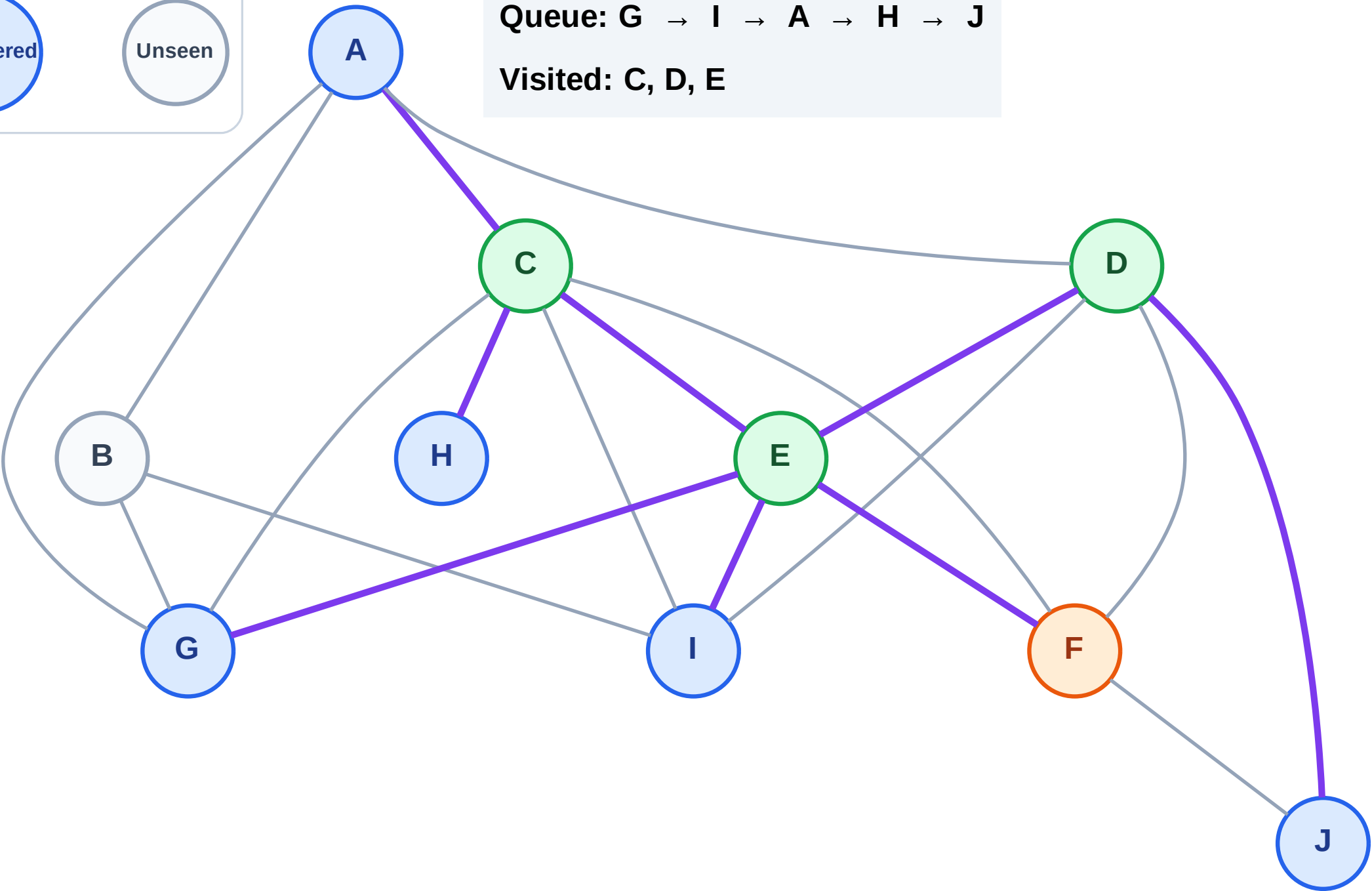
Processing

Discovered

Unseen

Queue: G → I → A → H → J

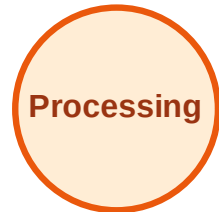
Visited: C, D, E



BFS Traversal

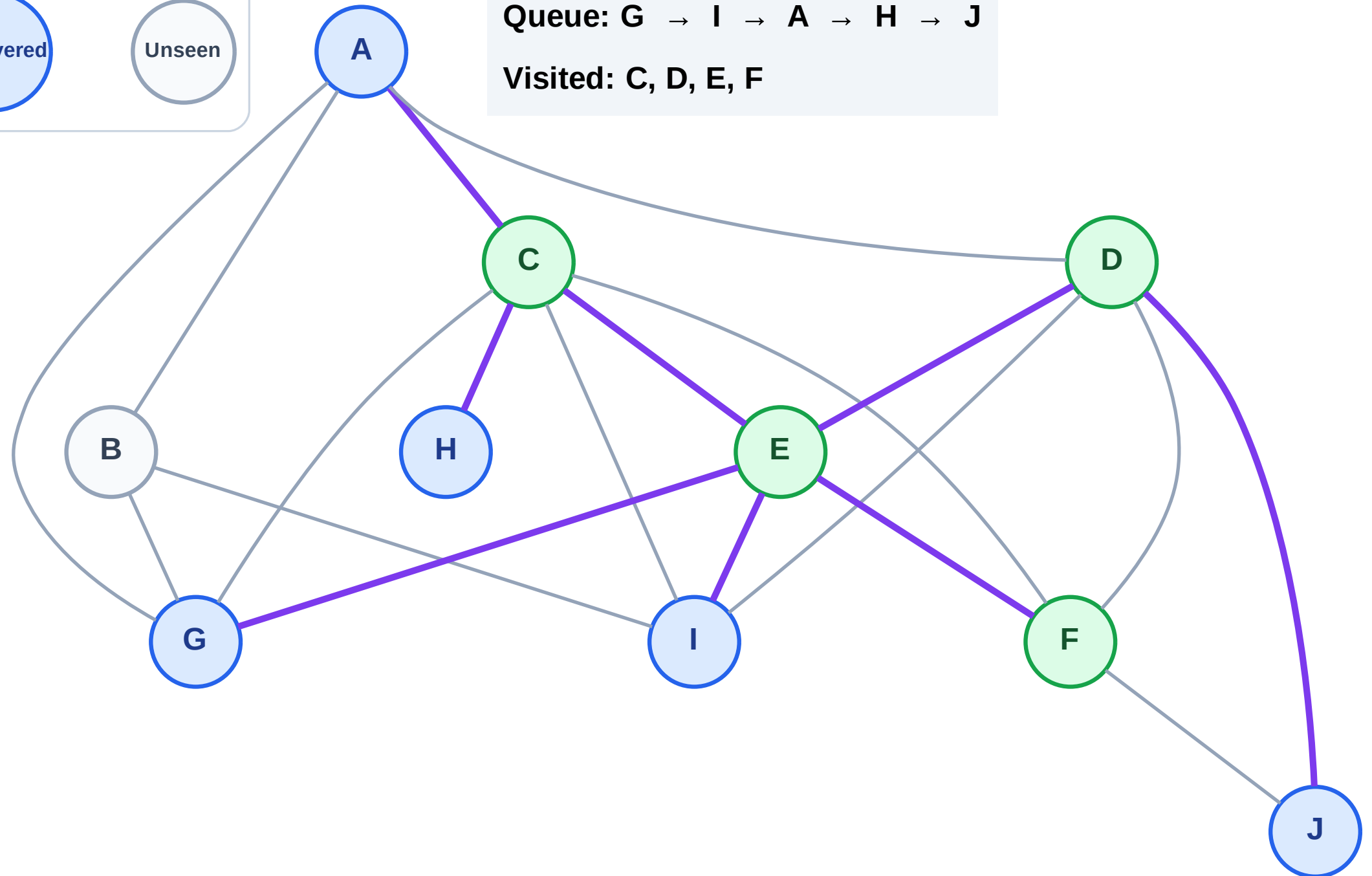
Step 9: No new neighbors from F

Legend



Queue: G → I → A → H → J

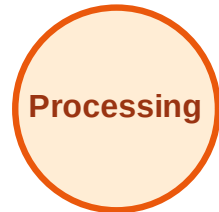
Visited: C, D, E, F



BFS Traversal

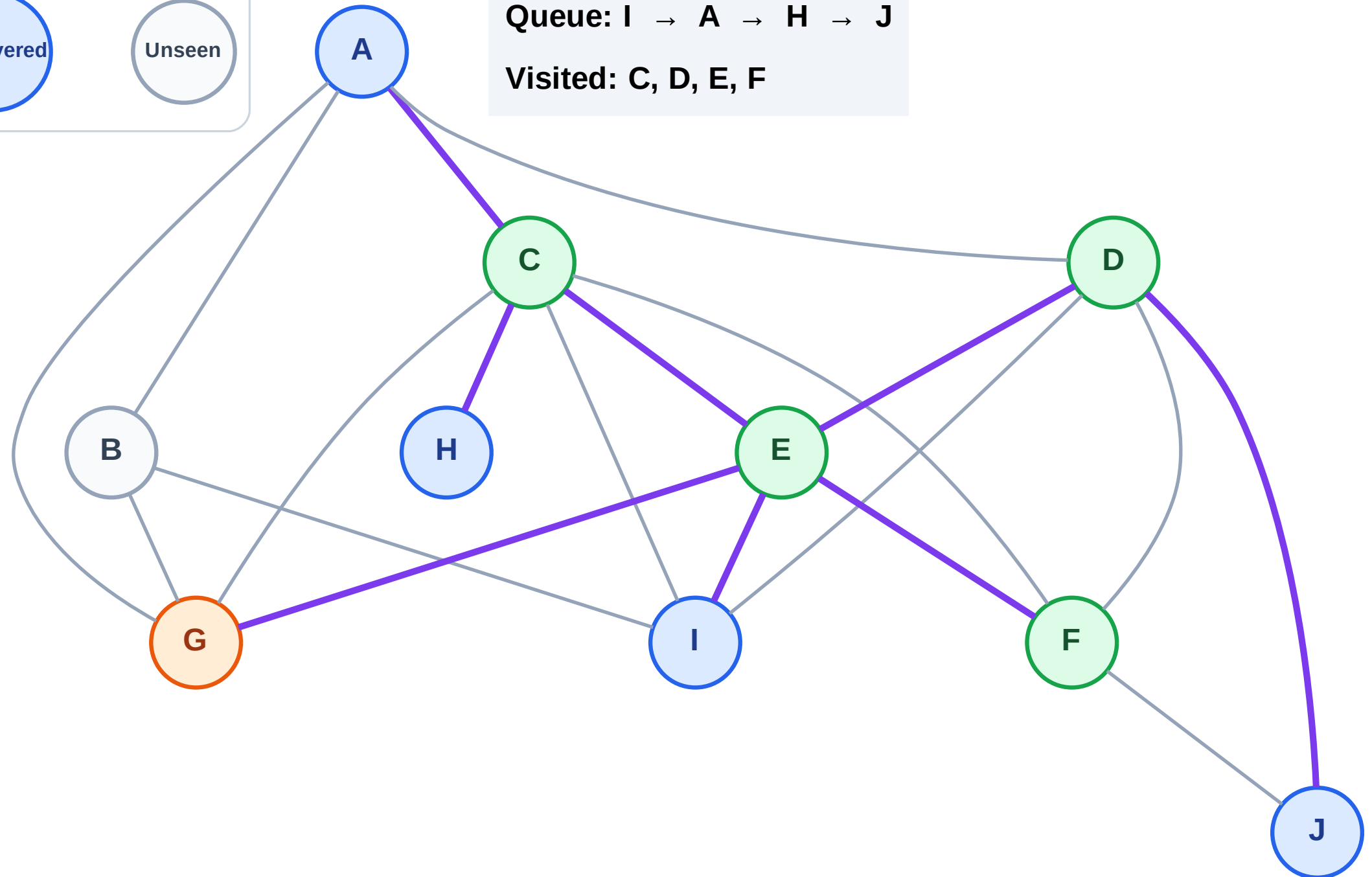
Step 10: Process node G

Legend



Queue: I → A → H → J

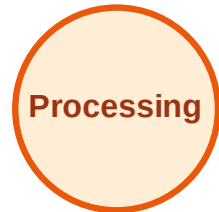
Visited: C, D, E, F



BFS Traversal

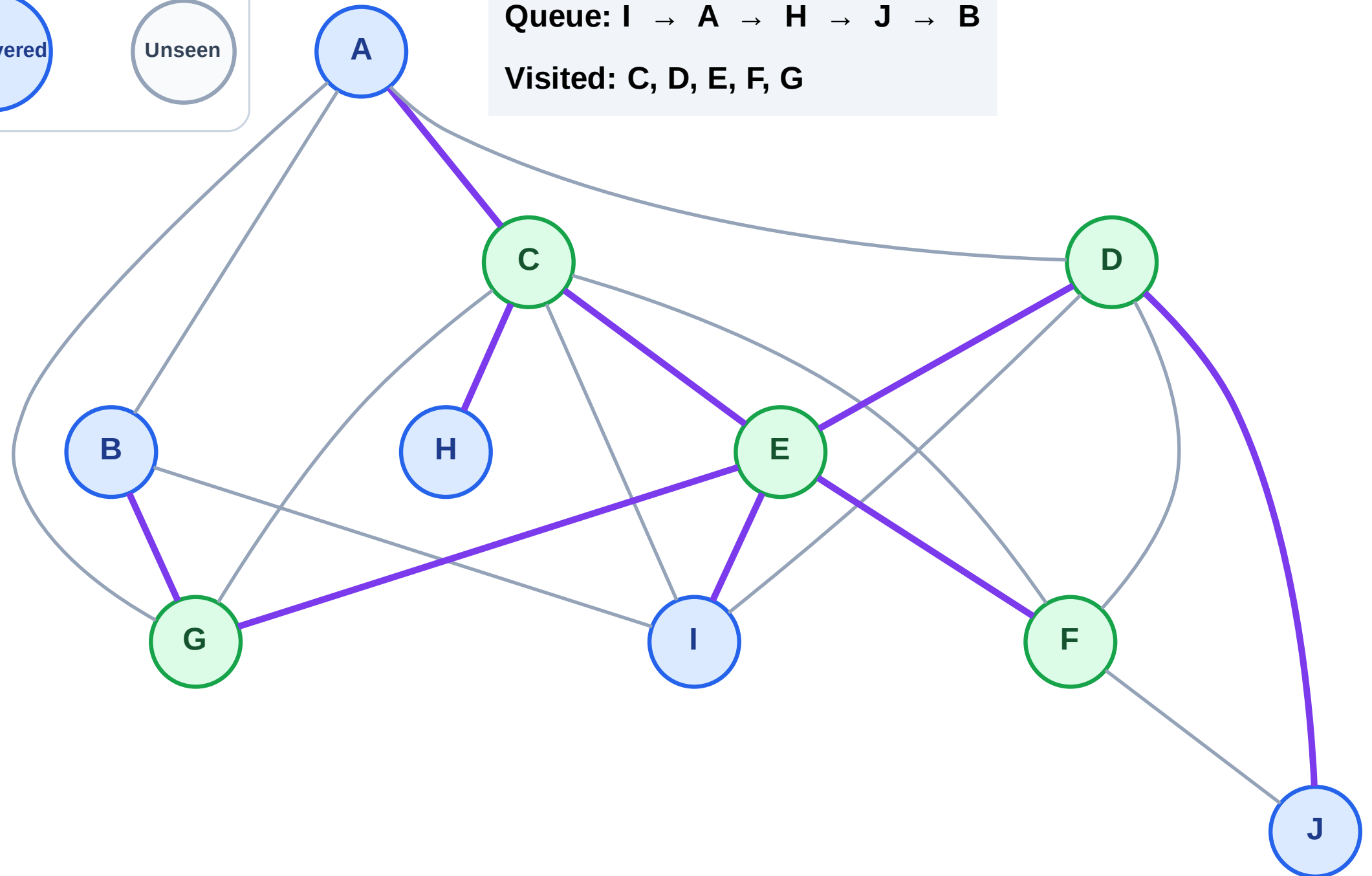
Step 11: Discover B from G

Legend



Queue: I → A → H → J → B

Visited: C, D, E, F, G



BFS Traversal

Step 12: Process node I

Legend

Complete

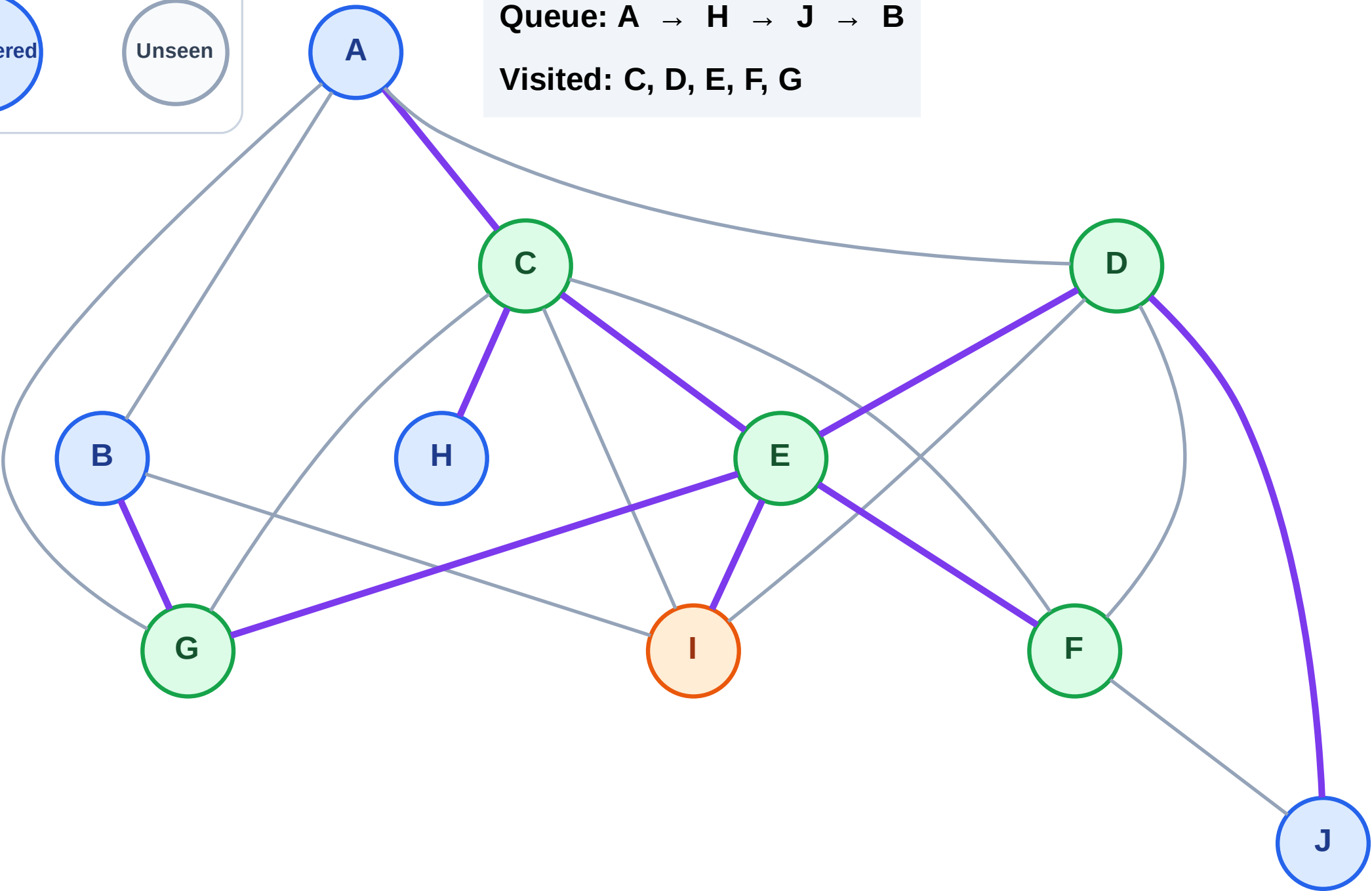
Processing

Discovered

Unseen

Queue: A → H → J → B

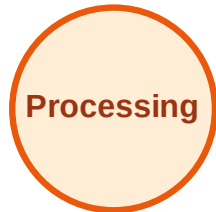
Visited: C, D, E, F, G



BFS Traversal

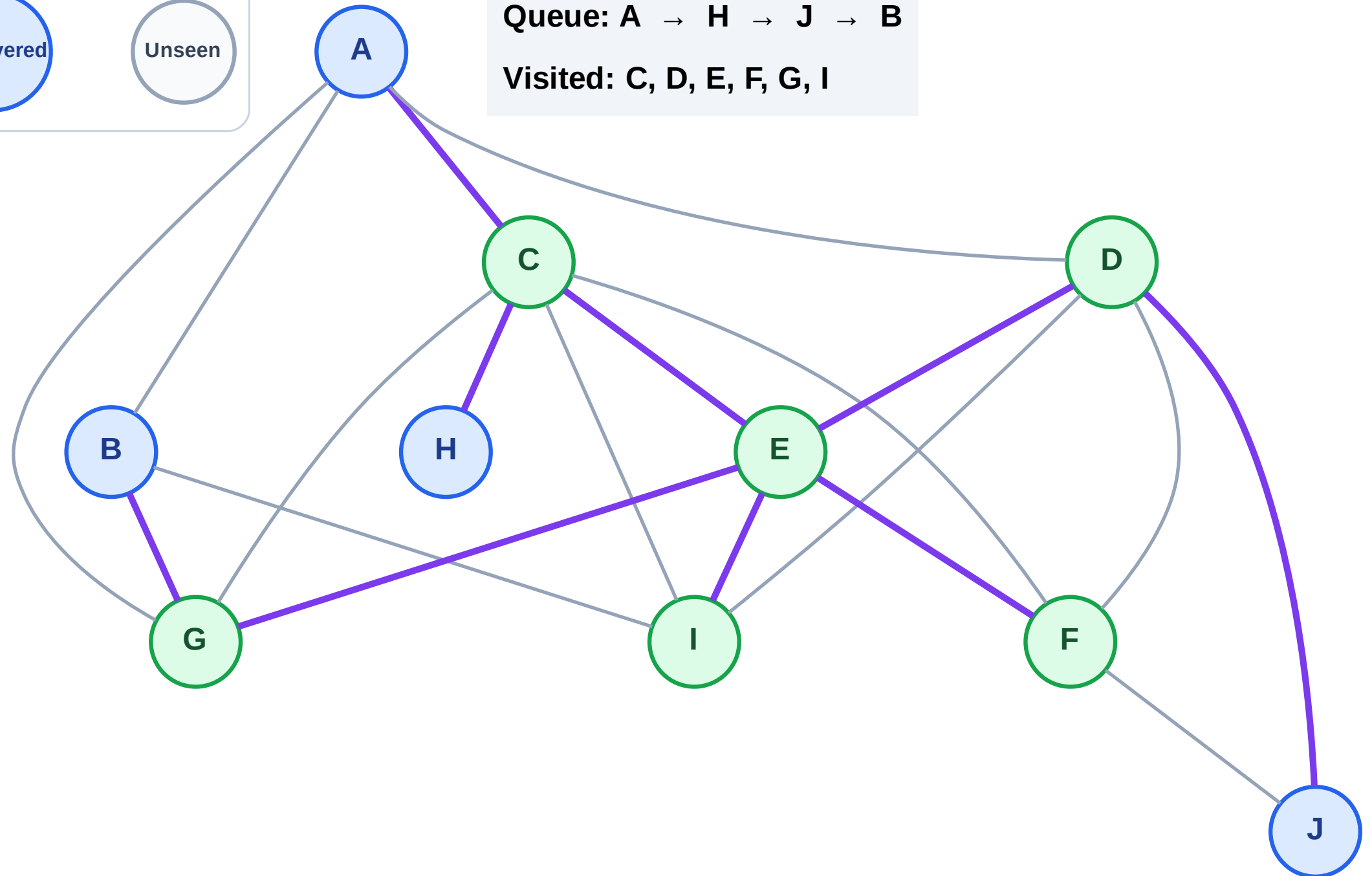
Step 13: No new neighbors from I

Legend



Queue: A → H → J → B

Visited: C, D, E, F, G, I



BFS Traversal

Step 14: Process node A

Legend

Complete

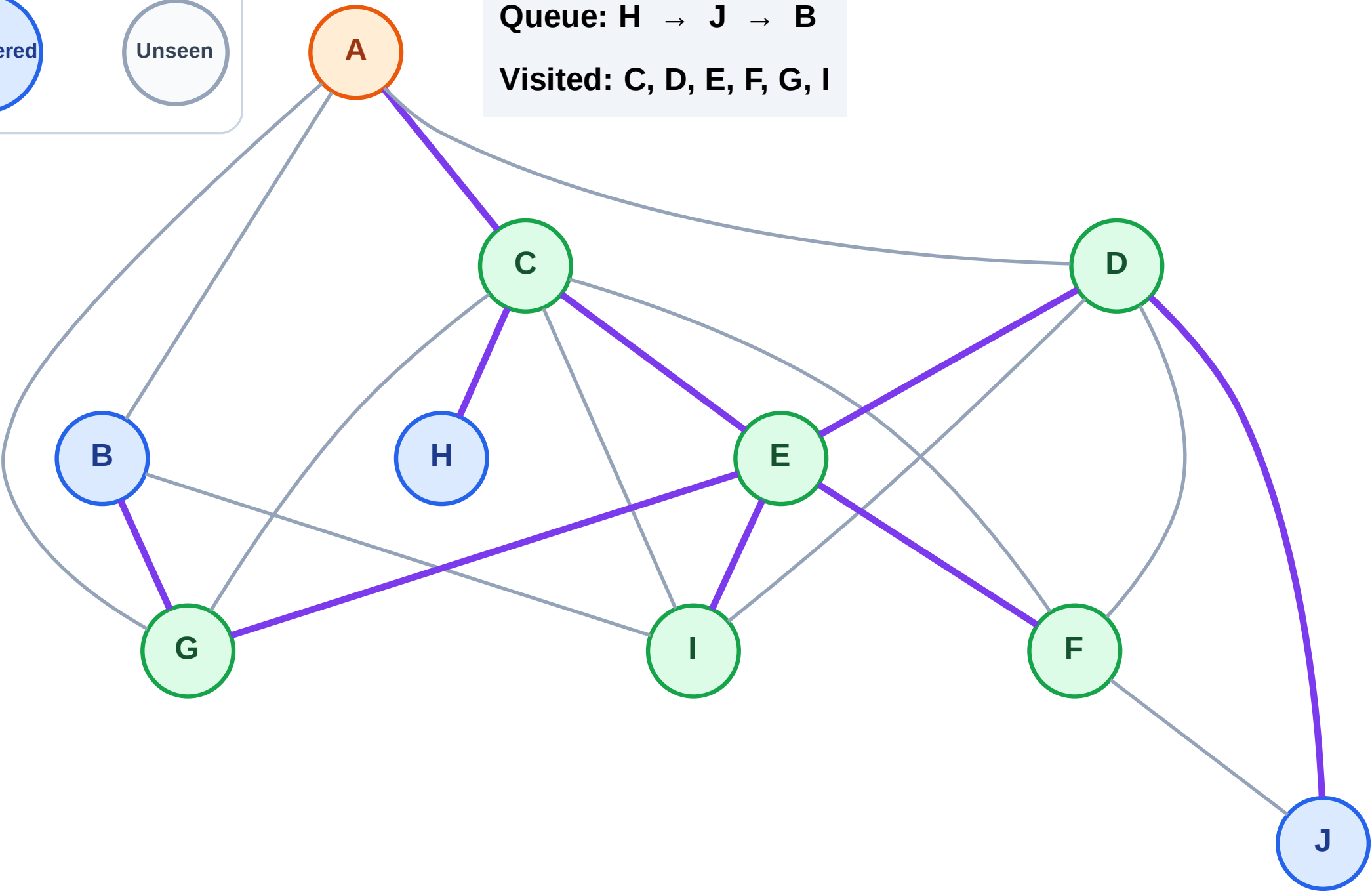
Processing

Discovered

Unseen

Queue: H → J → B

Visited: C, D, E, F, G, I



BFS Traversal

Step 15: No new neighbors from A

Legend

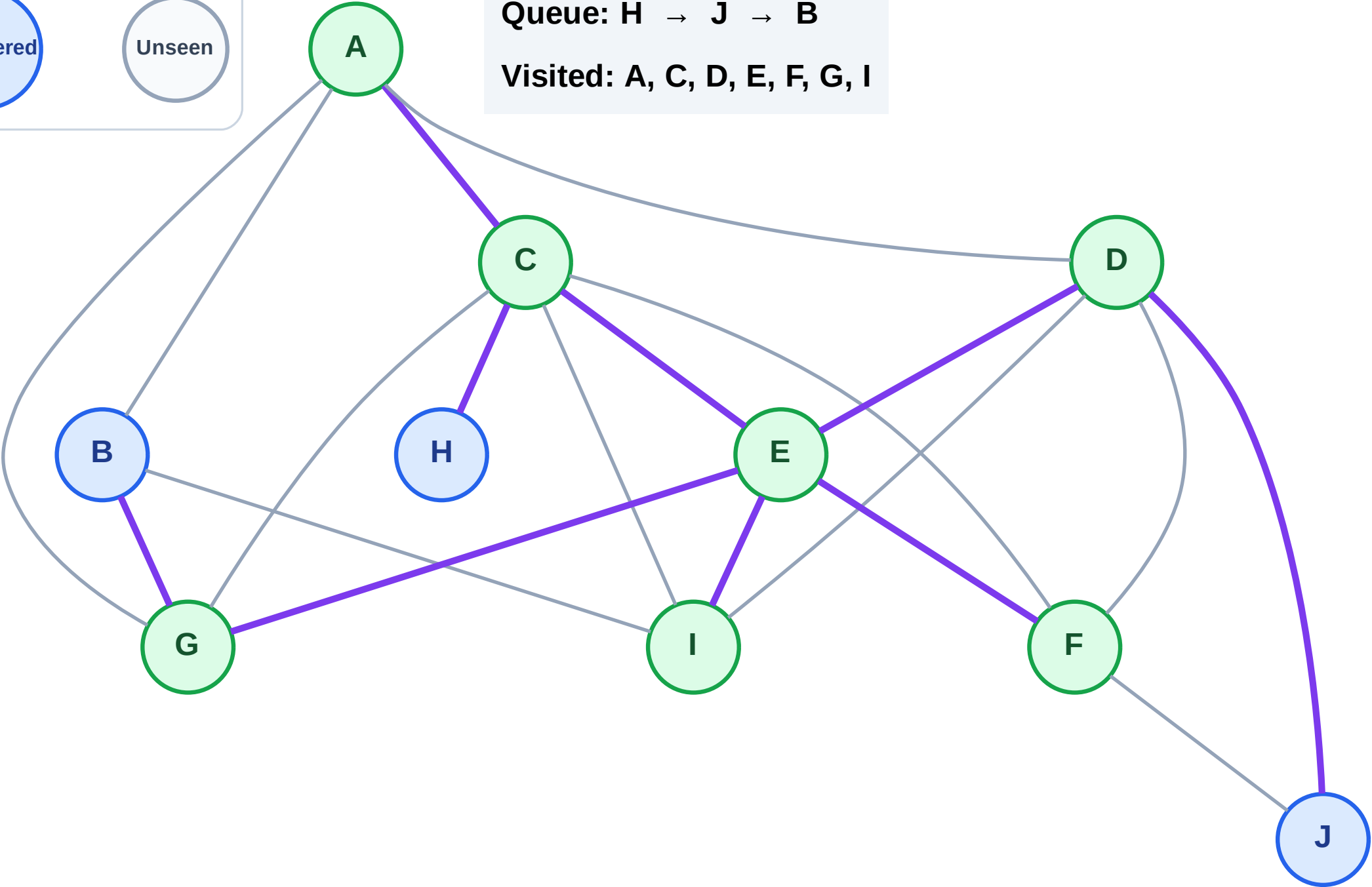
Complete

Processing

Discovered

Unseen

Queue: H → J → B
Visited: A, C, D, E, F, G, I



BFS Traversal

Step 16: Process node H

Legend

Complete

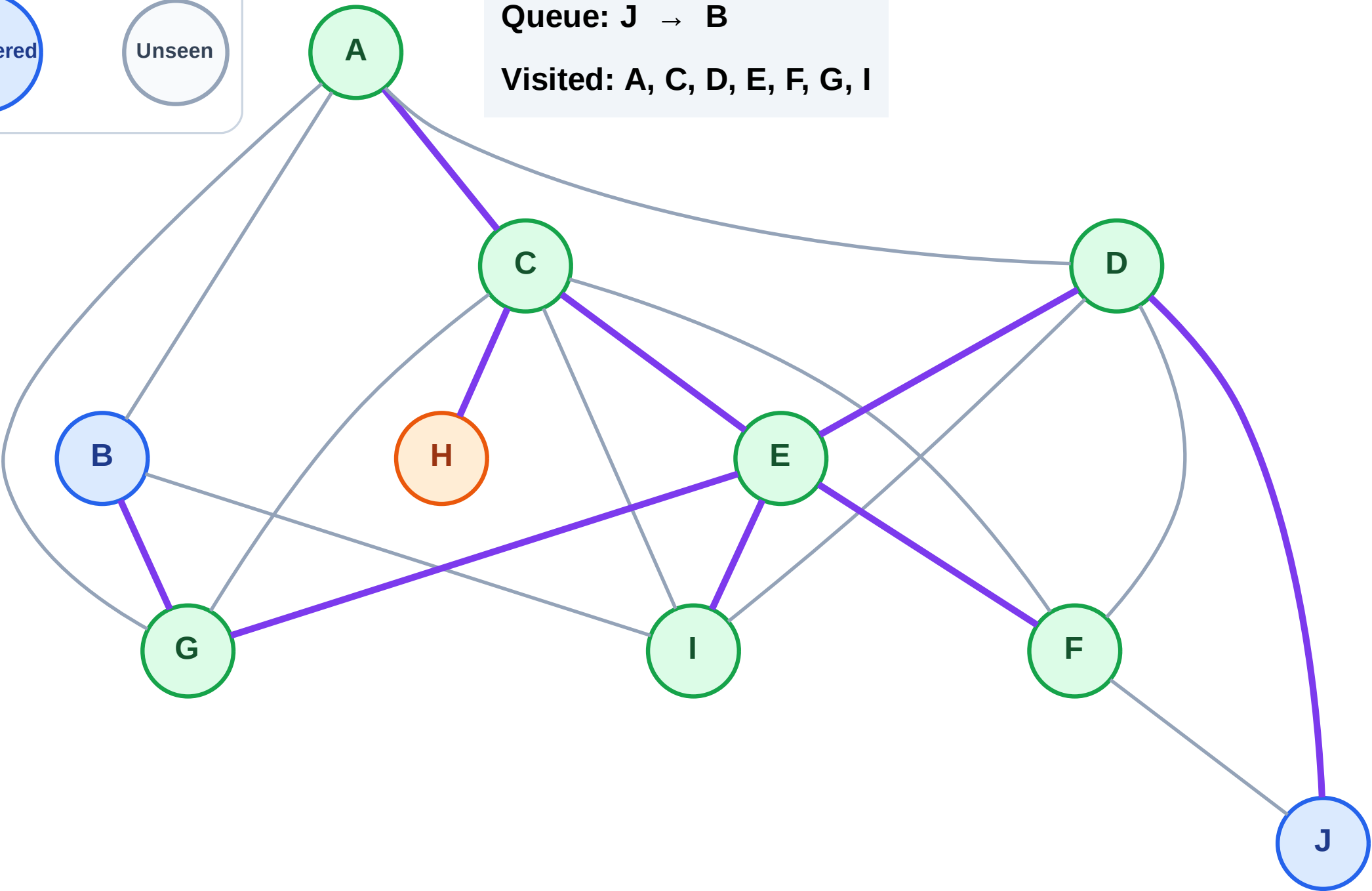
Processing

Discovered

Unseen

Queue: J → B

Visited: A, C, D, E, F, G, I



BFS Traversal

Step 17: No new neighbors from H

Legend

Complete

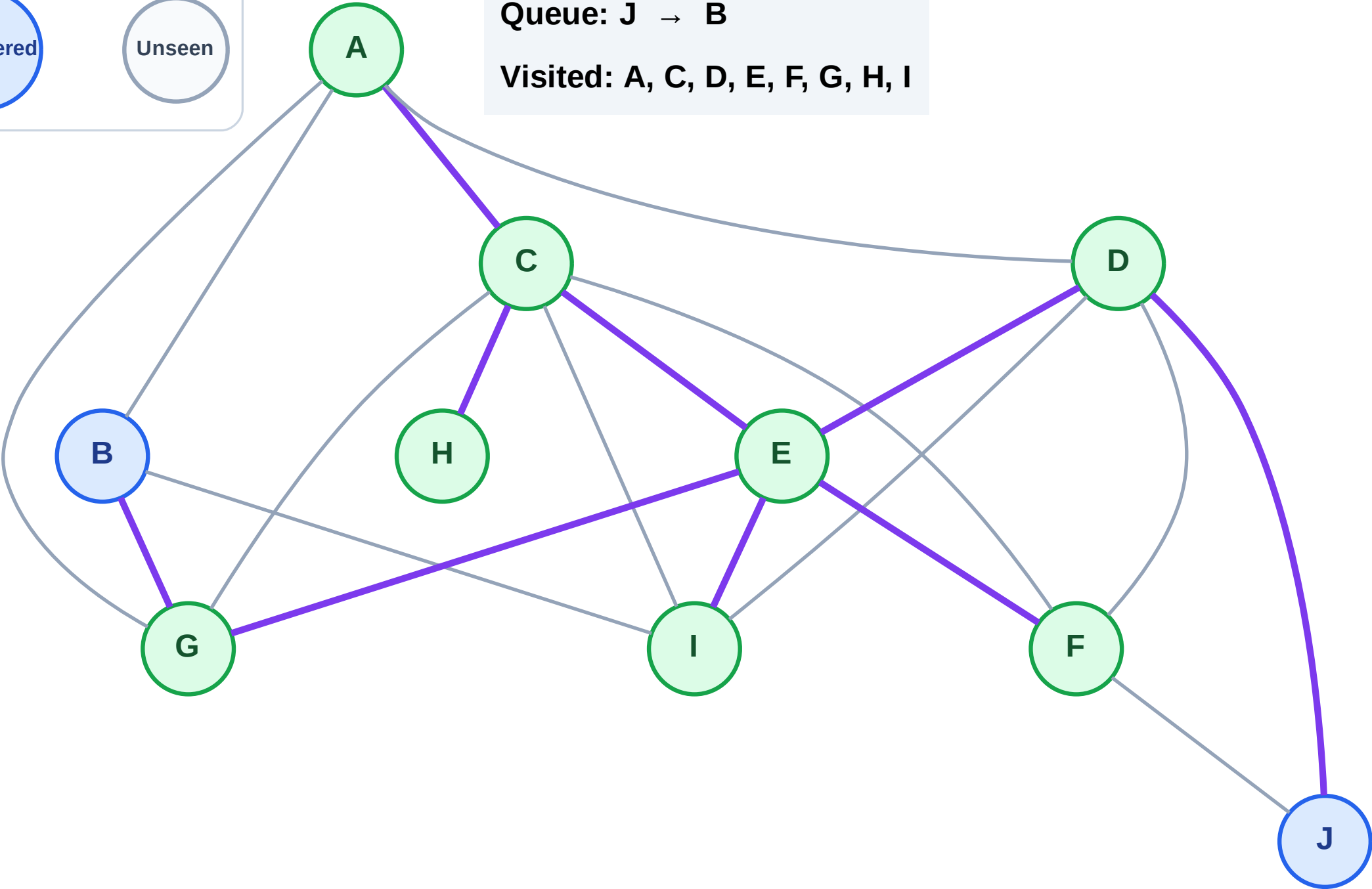
Processing

Discovered

Unseen

Queue: J → B

Visited: A, C, D, E, F, G, H, I



BFS Traversal

Step 18: Process node J

Legend

Complete

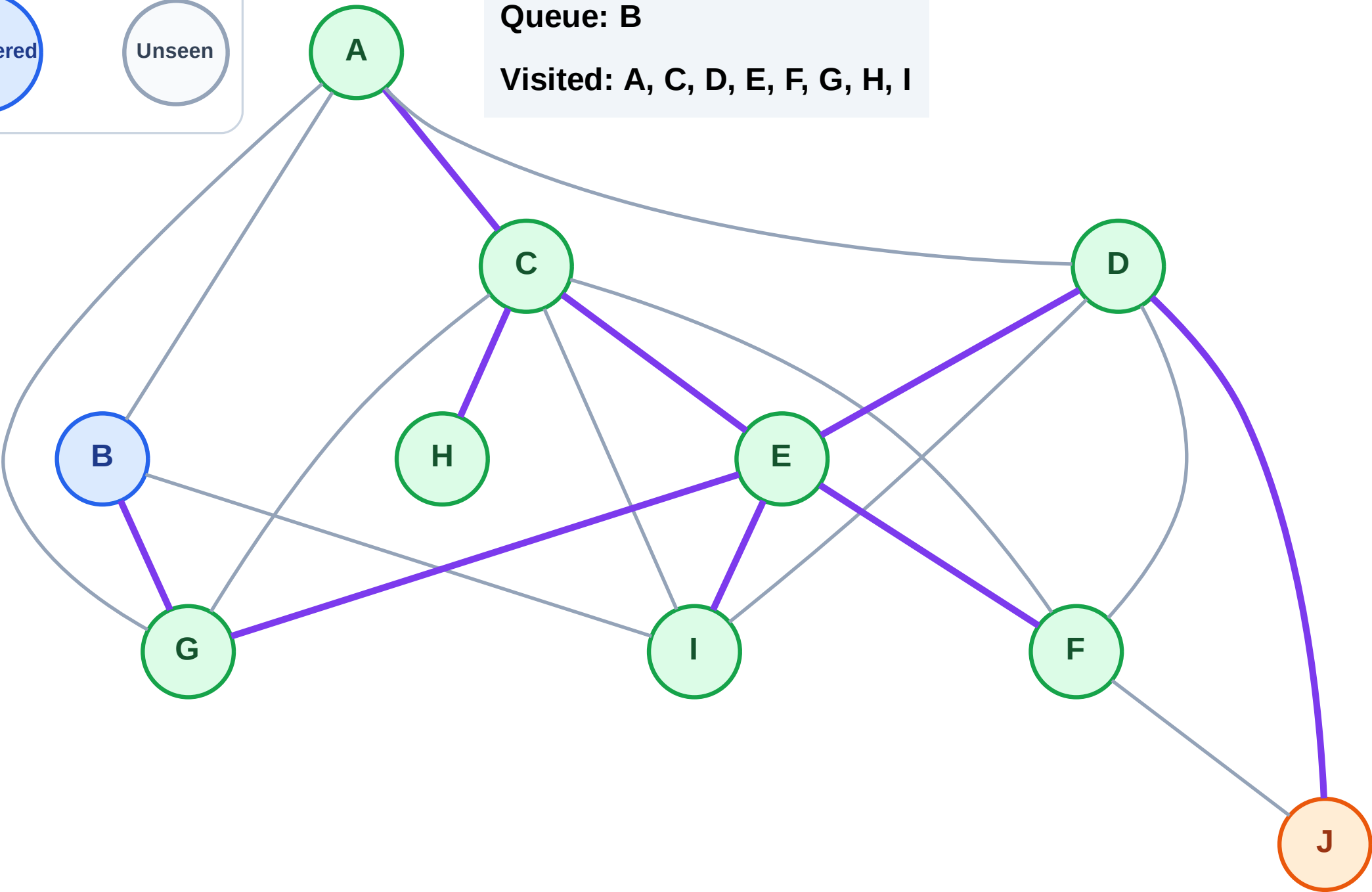
Processing

Discovered

Unseen

Queue: B

Visited: A, C, D, E, F, G, H, I



BFS Traversal

Step 19: No new neighbors from J

Legend

Complete

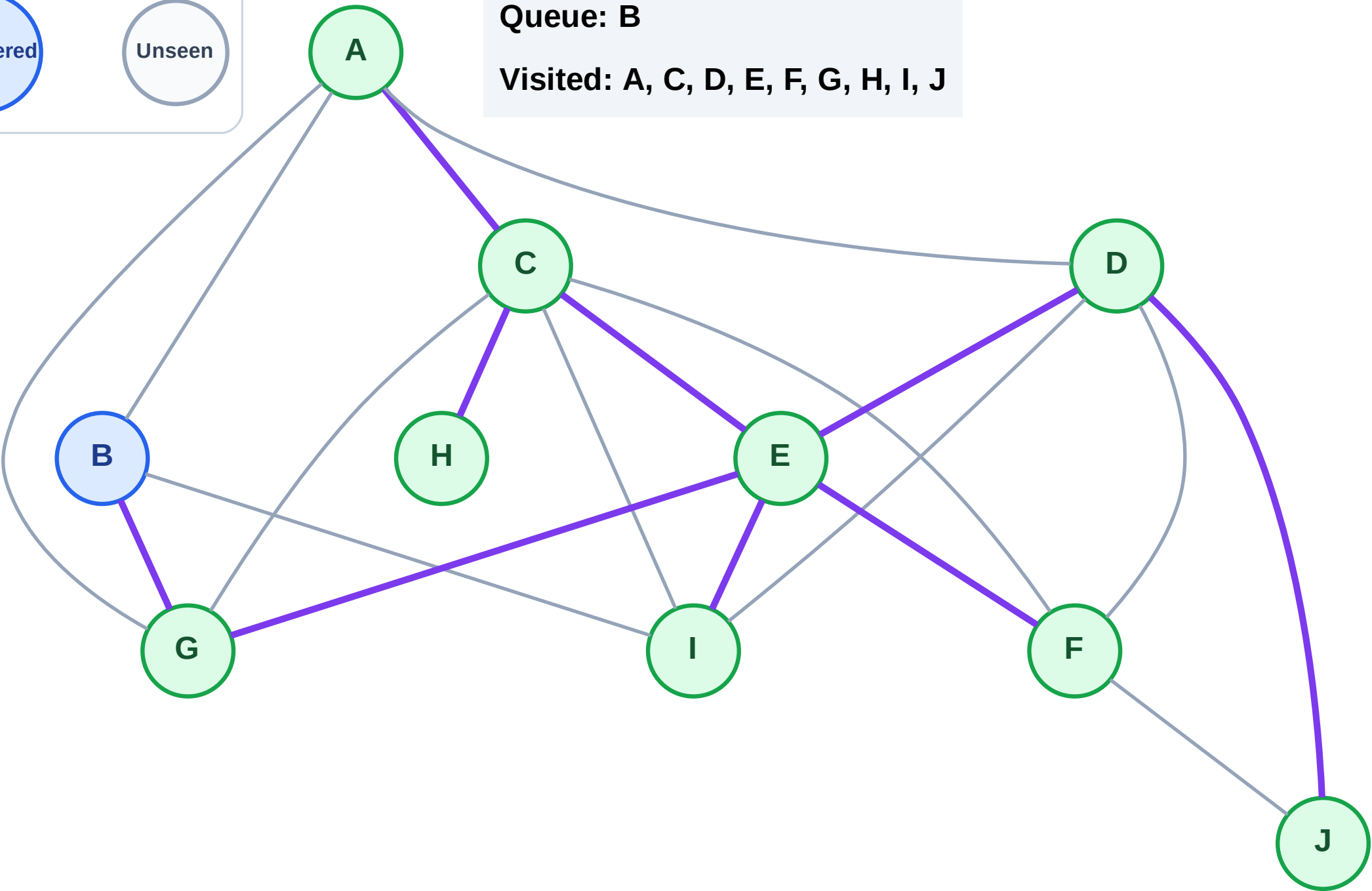
Processing

Discovered

Unseen

Queue: B

Visited: A, C, D, E, F, G, H, I, J



BFS Traversal

Step 20: Process node B

Legend

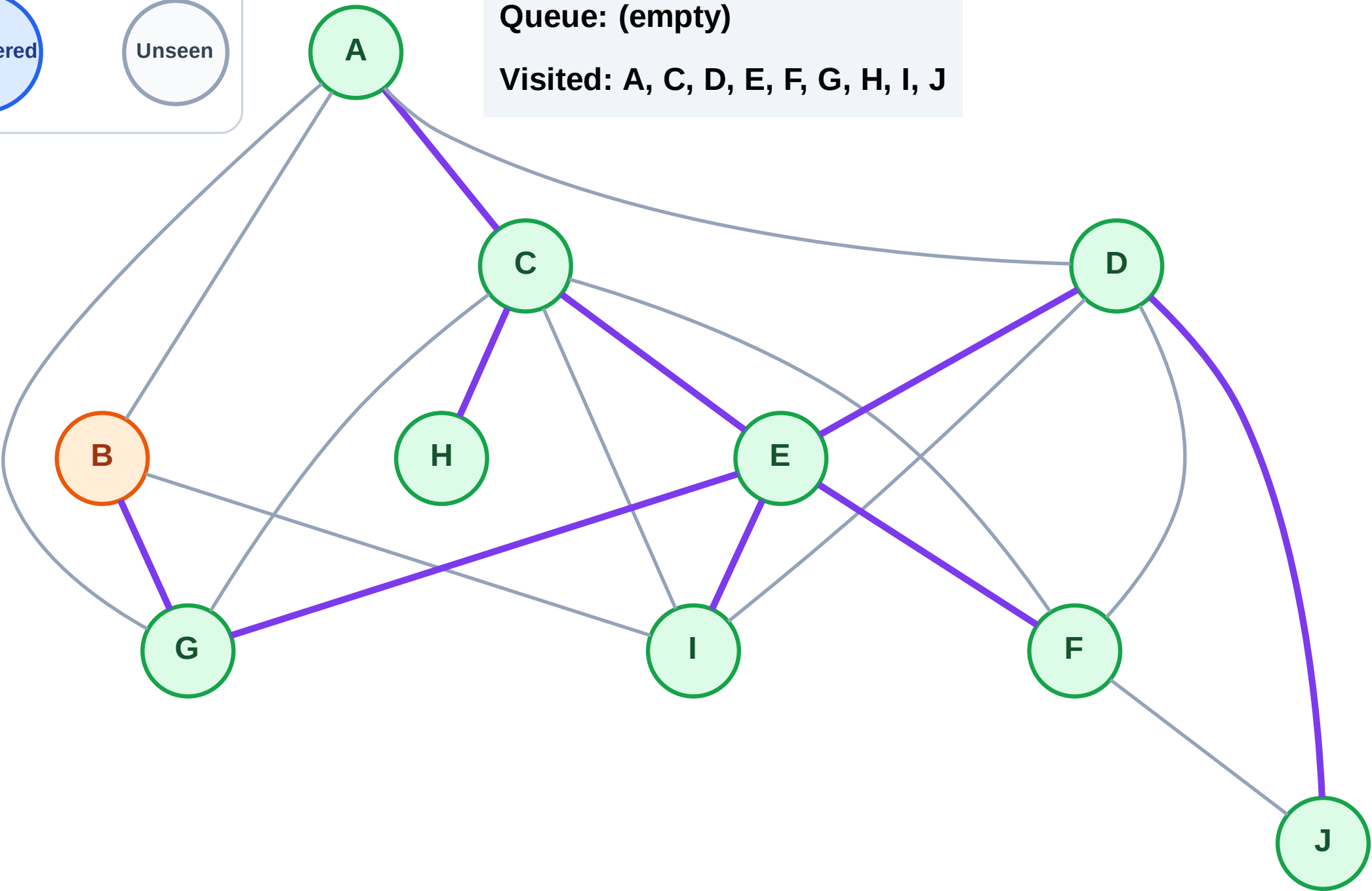
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: A, C, D, E, F, G, H, I, J



BFS Traversal

Step 21: No new neighbors from B

Legend

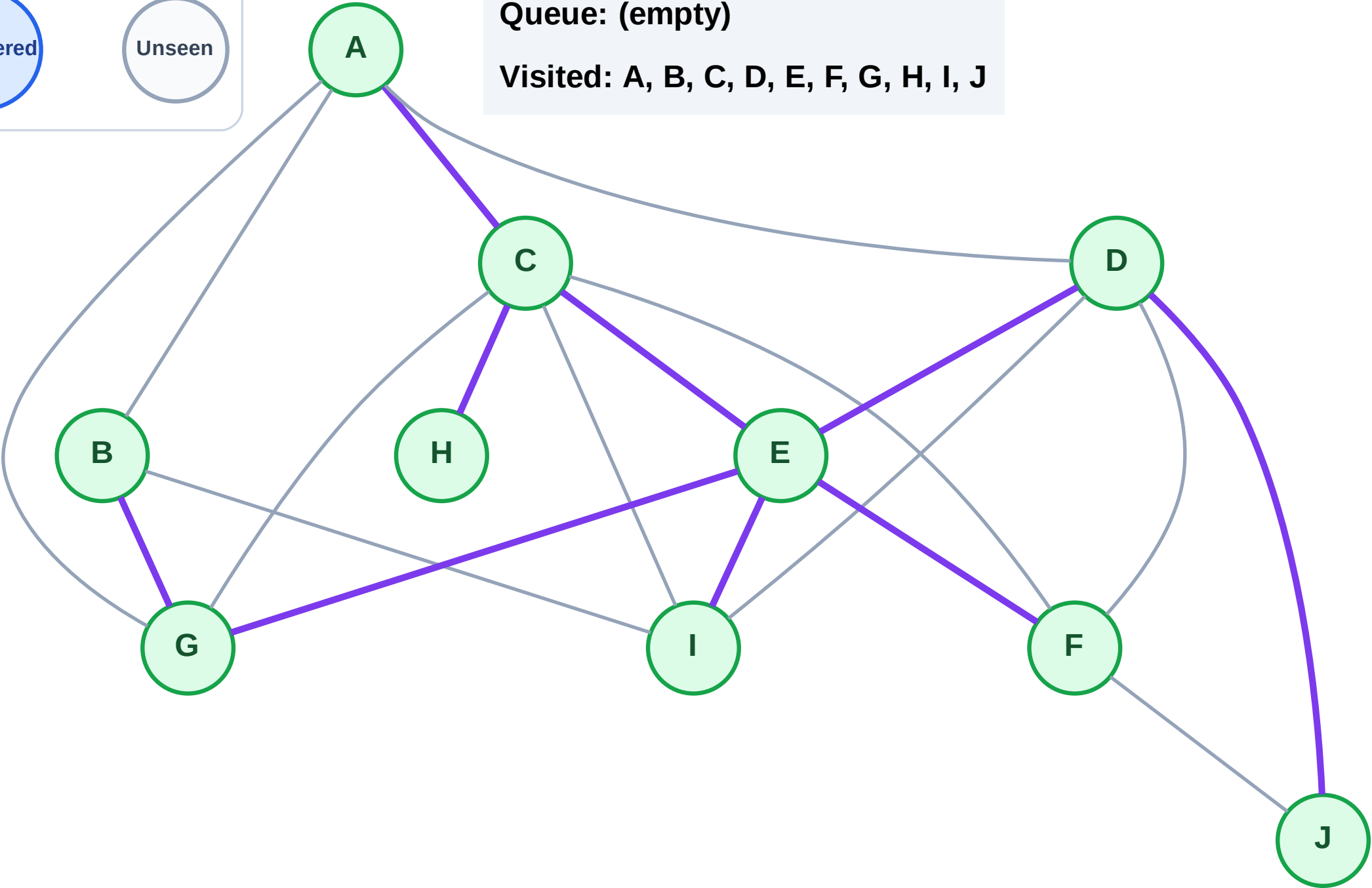
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at E

Legend

Complete

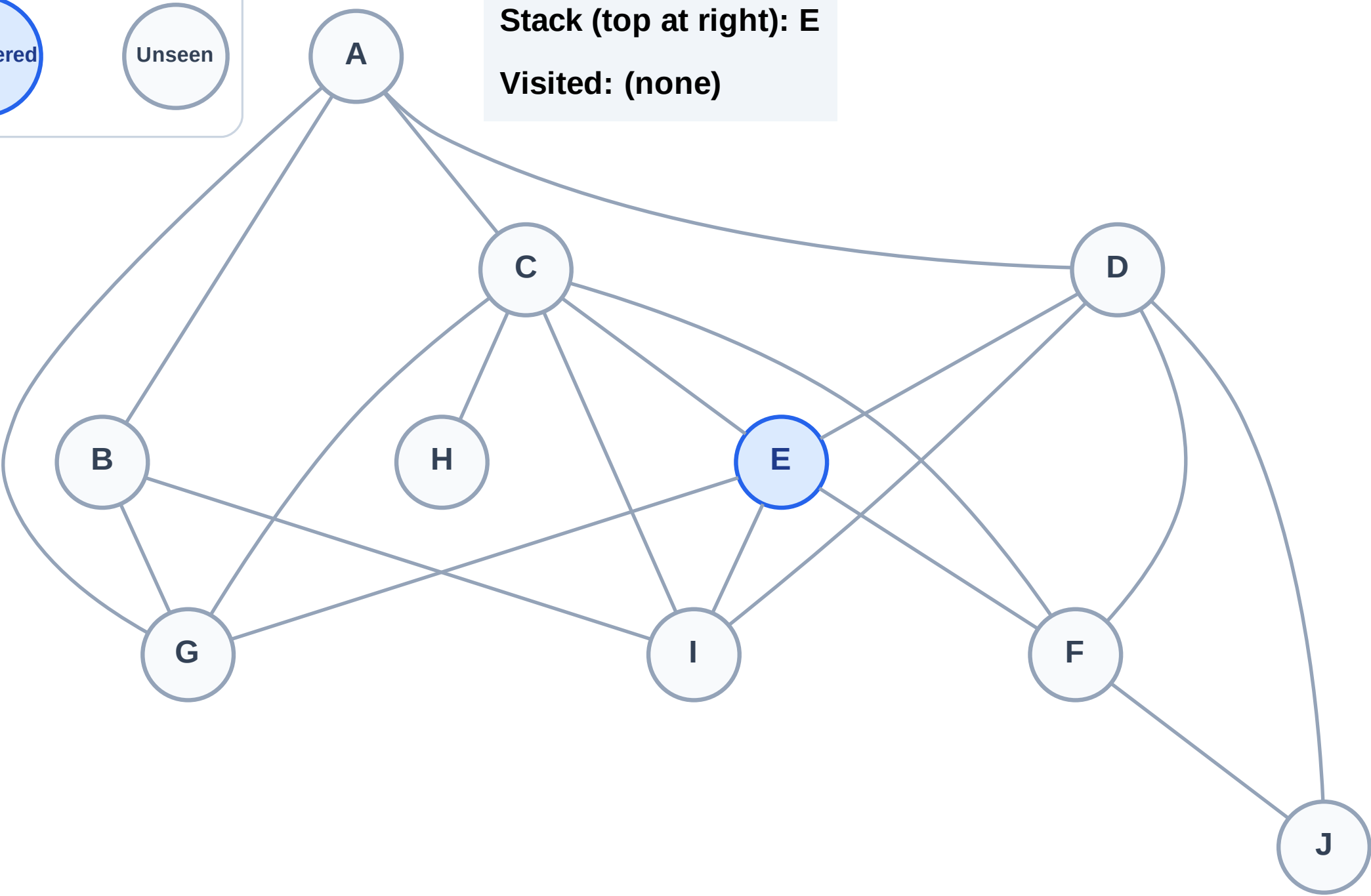
Processing

Discovered

Unseen

Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

Complete

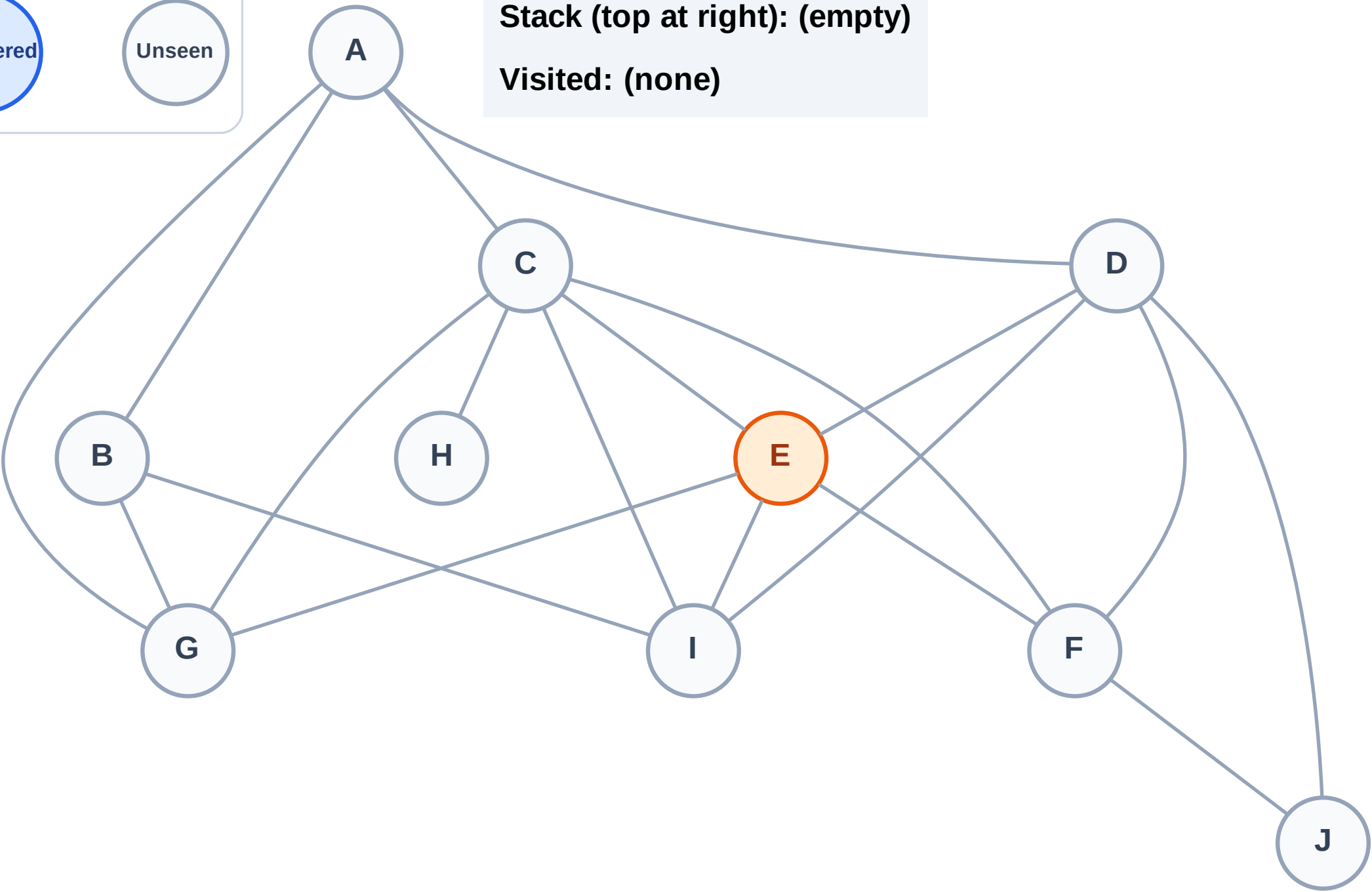
Processing

Discovered

Unseen

Stack (top at right): (empty)

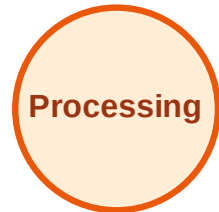
Visited: (none)



DFS Traversal

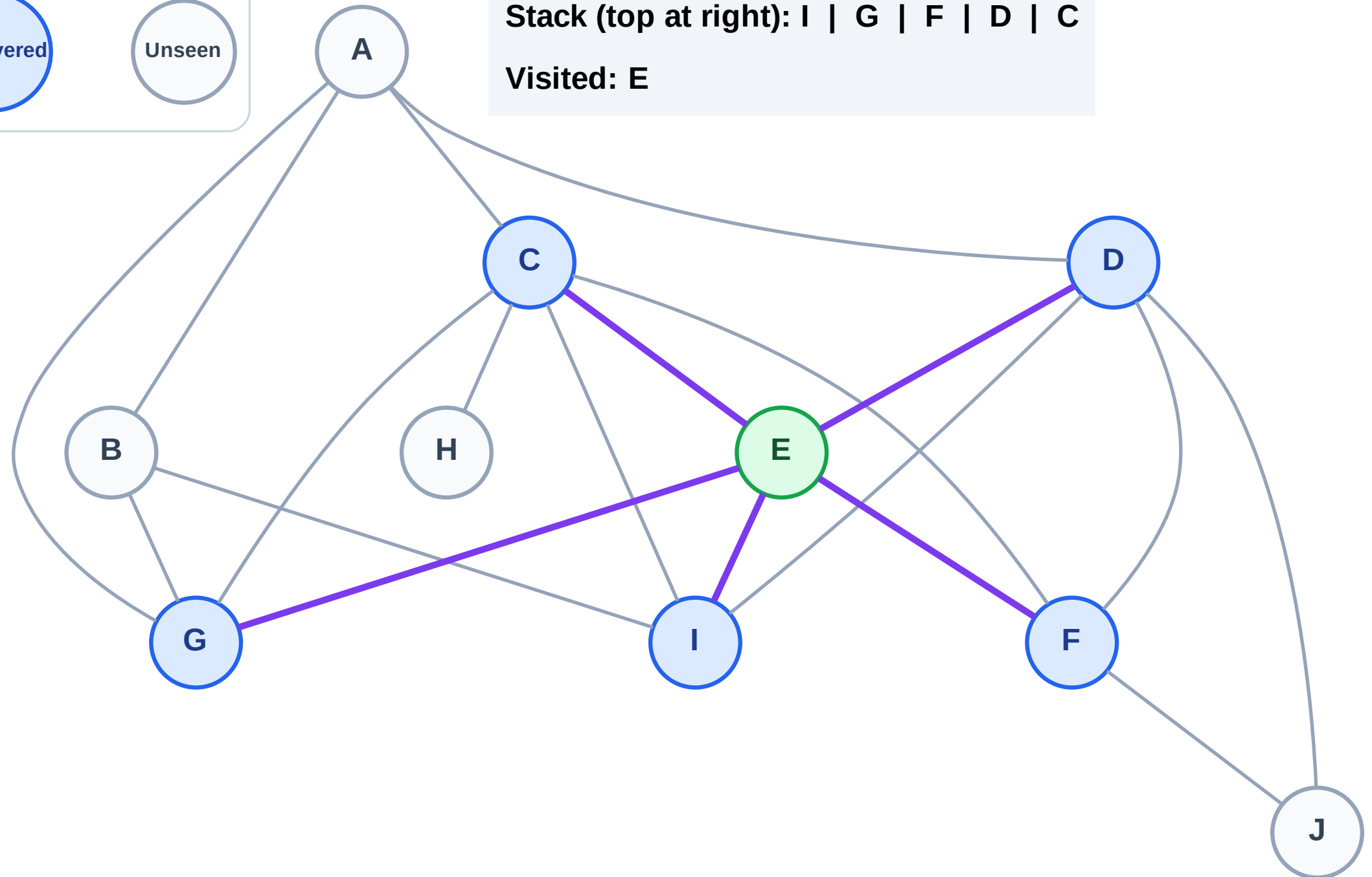
Step 3: Discover I, G, F, D, C from E

Legend



Stack (top at right): I | G | F | D | C

Visited: E



DFS Traversal

Step 4: Process node C

Legend

Complete

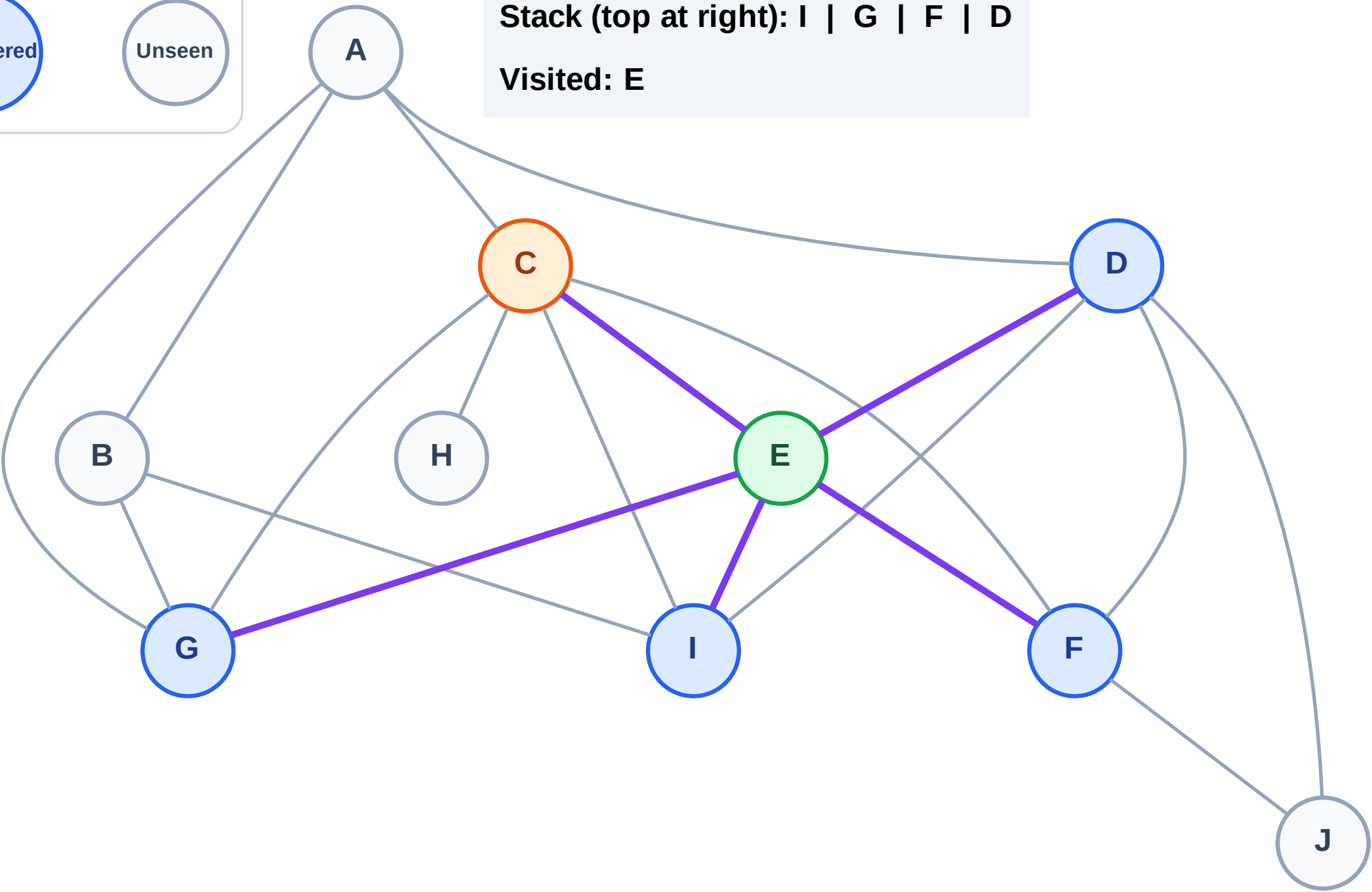
Processing

Discovered

Unseen

Stack (top at right): I | G | F | D

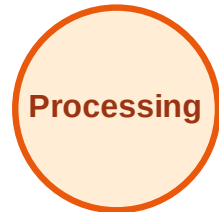
Visited: E



DFS Traversal

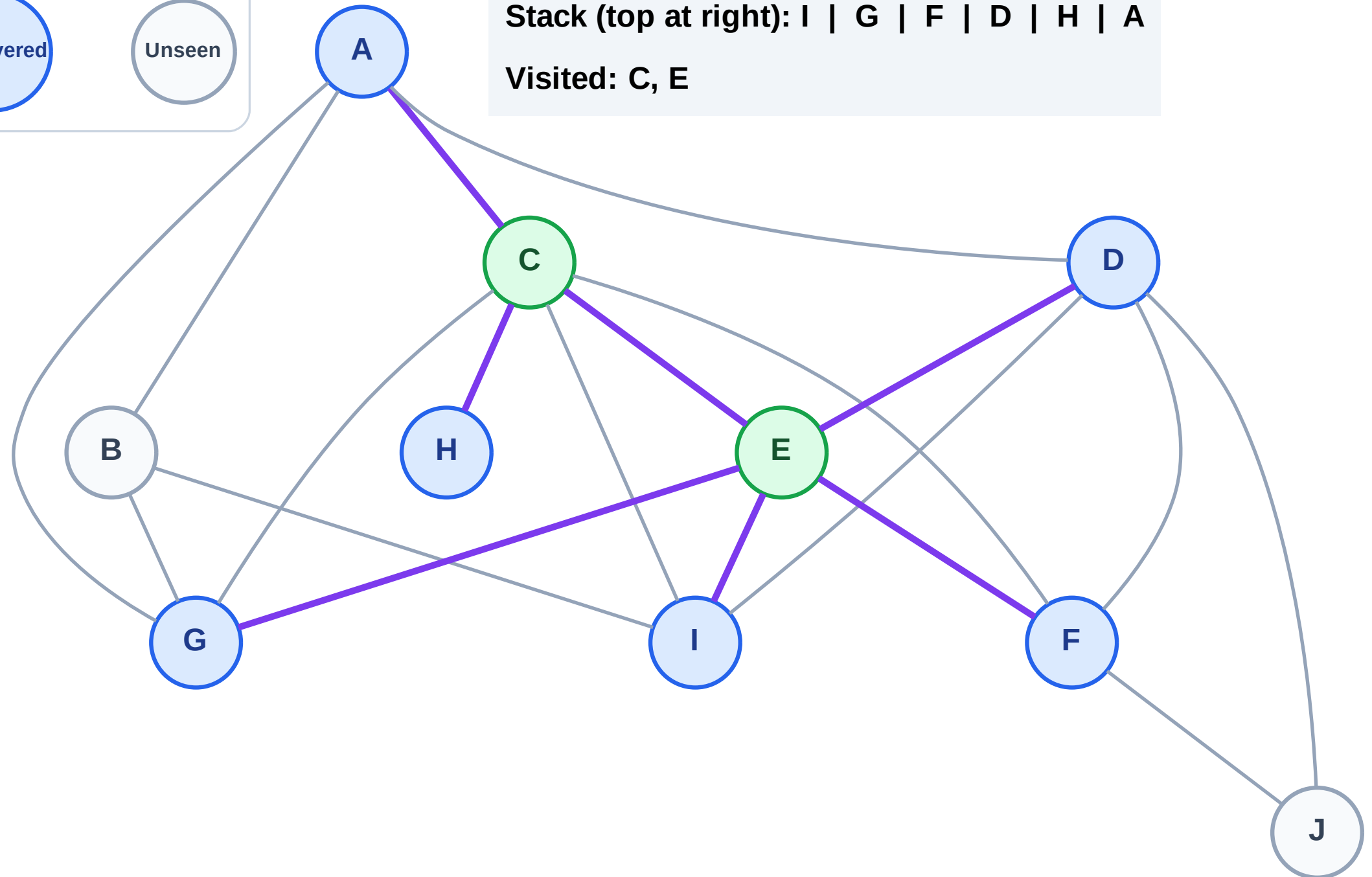
Step 5: Discover H, A from C

Legend



Stack (top at right): I | G | F | D | H | A

Visited: C, E



DFS Traversal

Step 6: Process node A

Legend

Complete

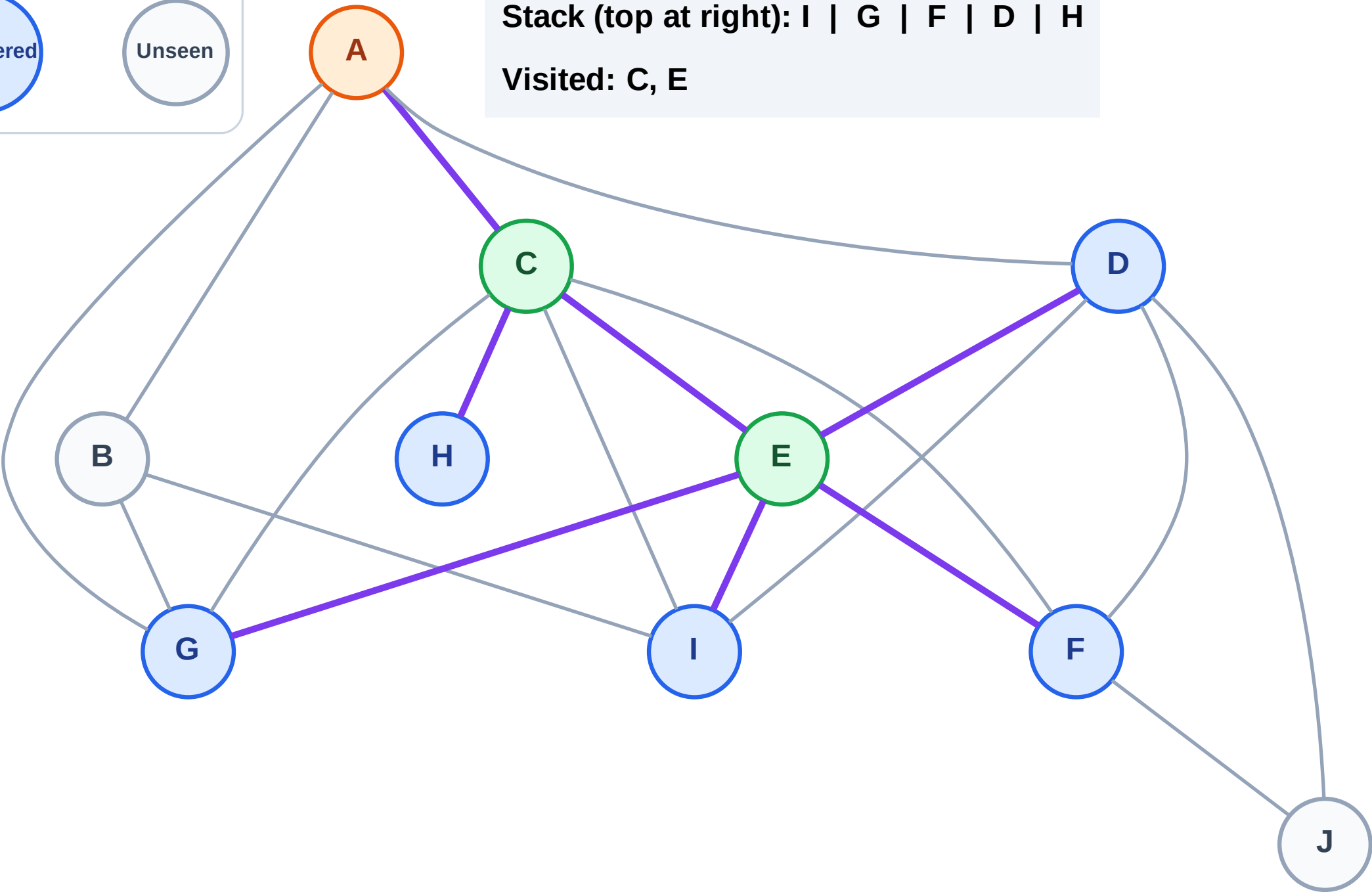
Processing

Discovered

Unseen

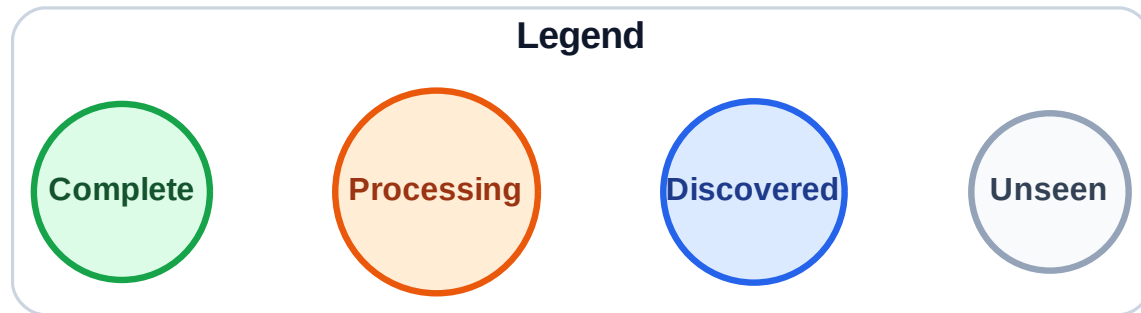
Stack (top at right): I | G | F | D | H

Visited: C, E



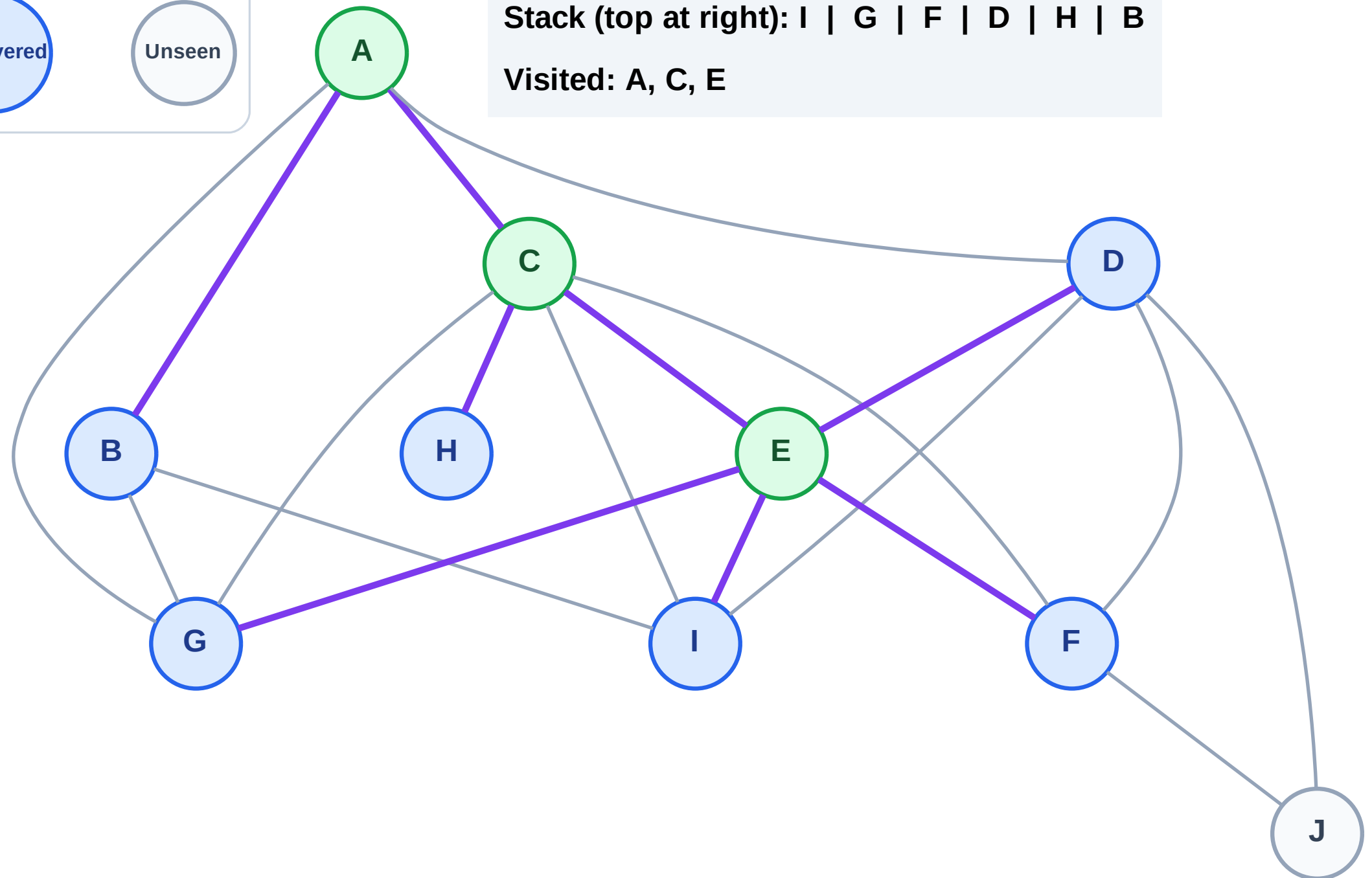
Step 7: Discover B from A

Step 7: Discover B from A



Stack (top at right): I | G | F | D | H | B

Visited: A, C, E



DFS Traversal

Step 8: Process node B

Legend

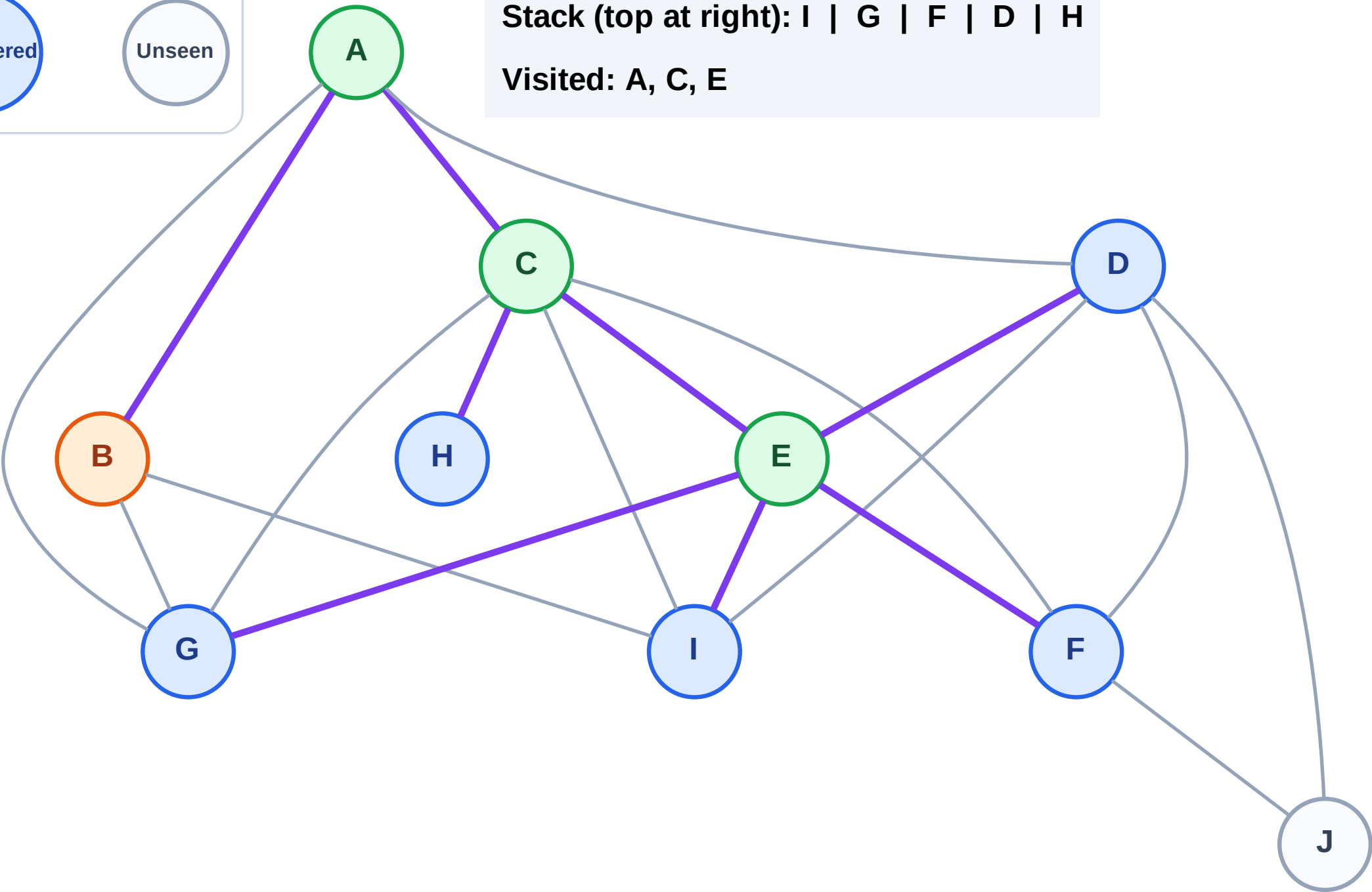
Complete

Processing

Discovered

Unseen

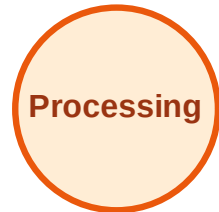
Stack (top at right): I | G | F | D | H
Visited: A, C, E



DFS Traversal

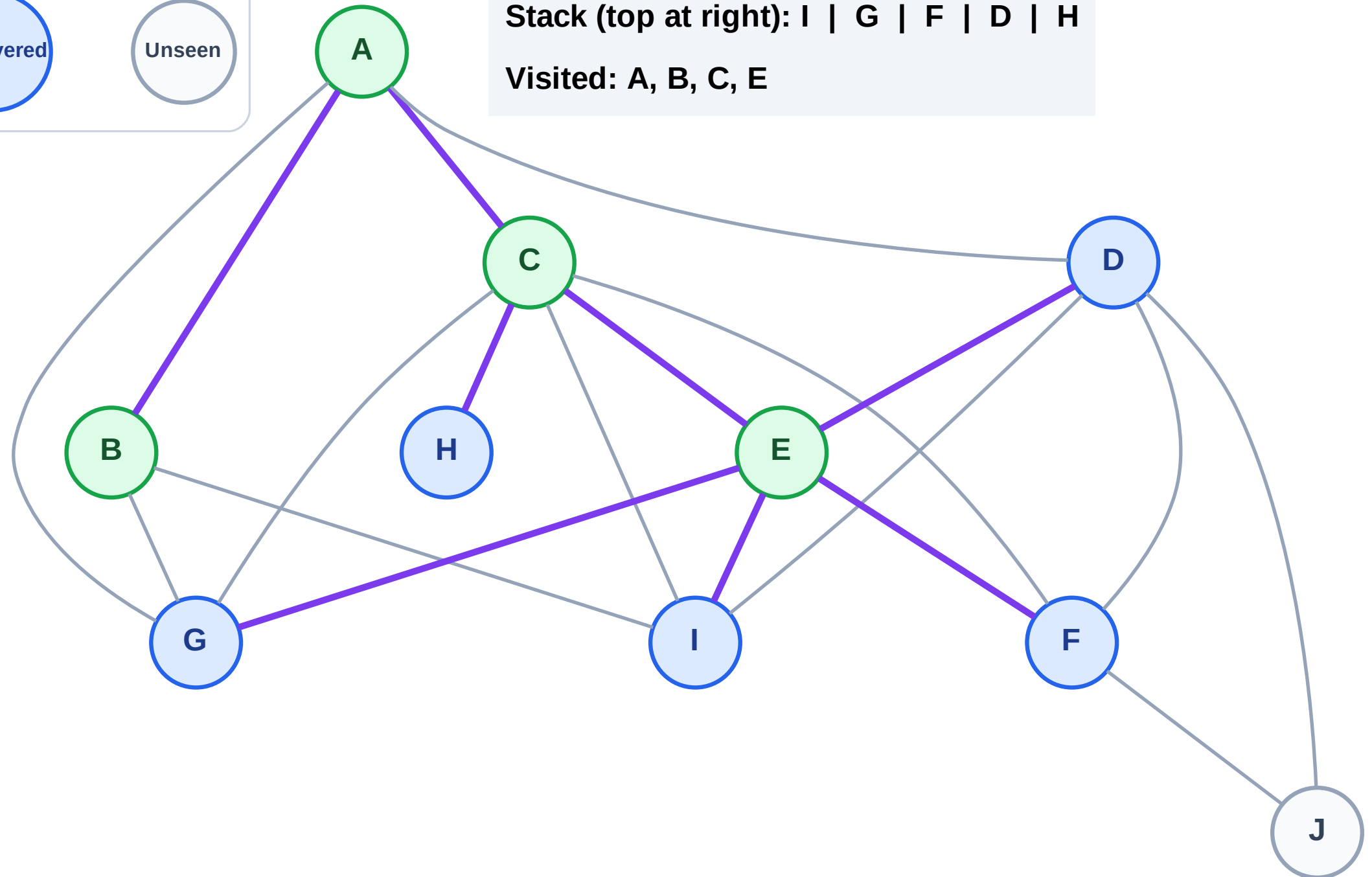
Step 9: No new neighbors from B

Legend



Stack (top at right): I | G | F | D | H

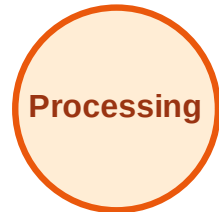
Visited: A, B, C, E



DFS Traversal

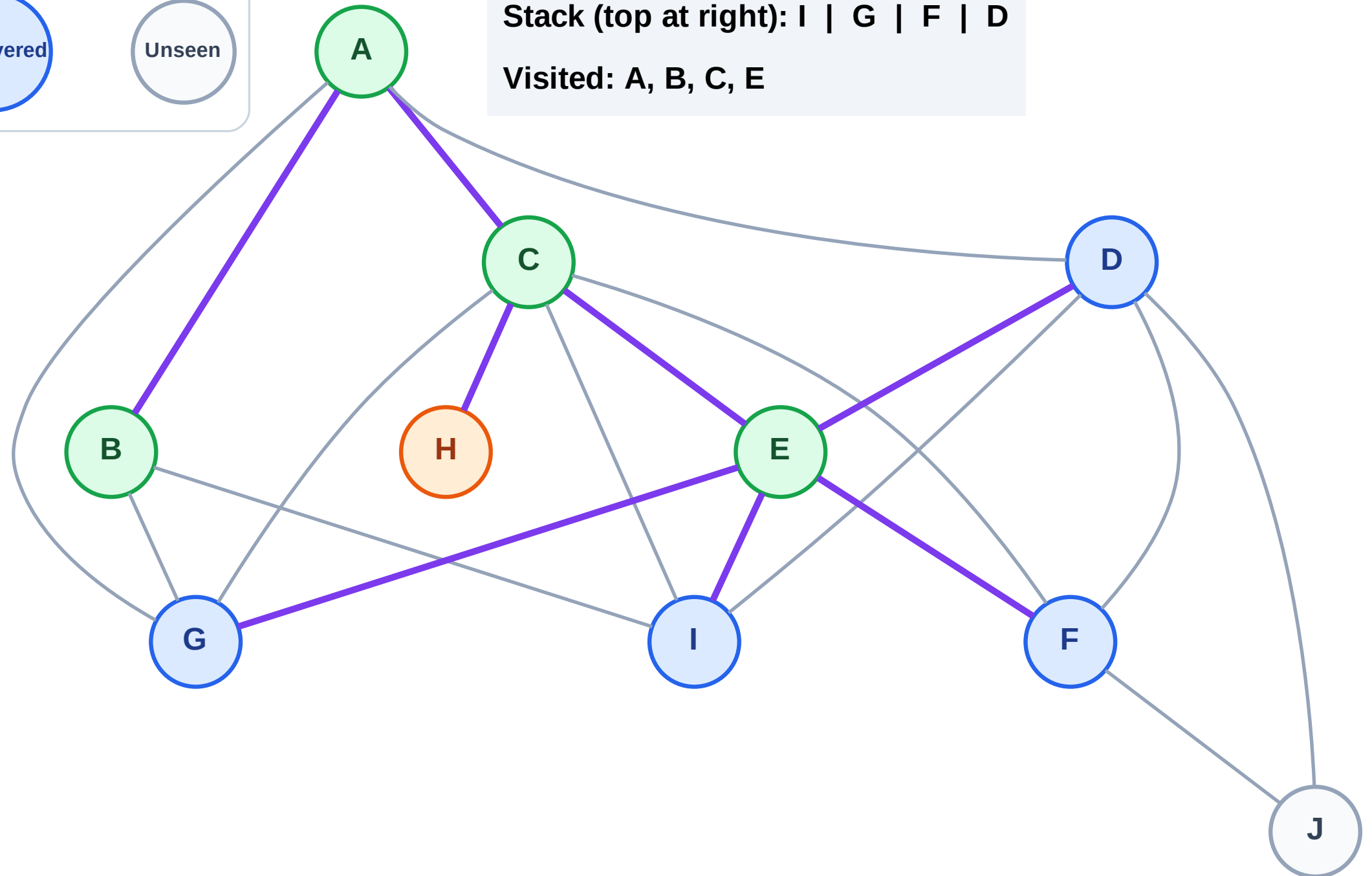
Step 10: Process node H

Legend



Stack (top at right): I | G | F | D

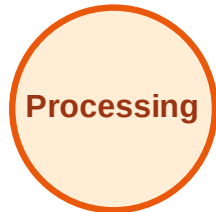
Visited: A, B, C, E



DFS Traversal

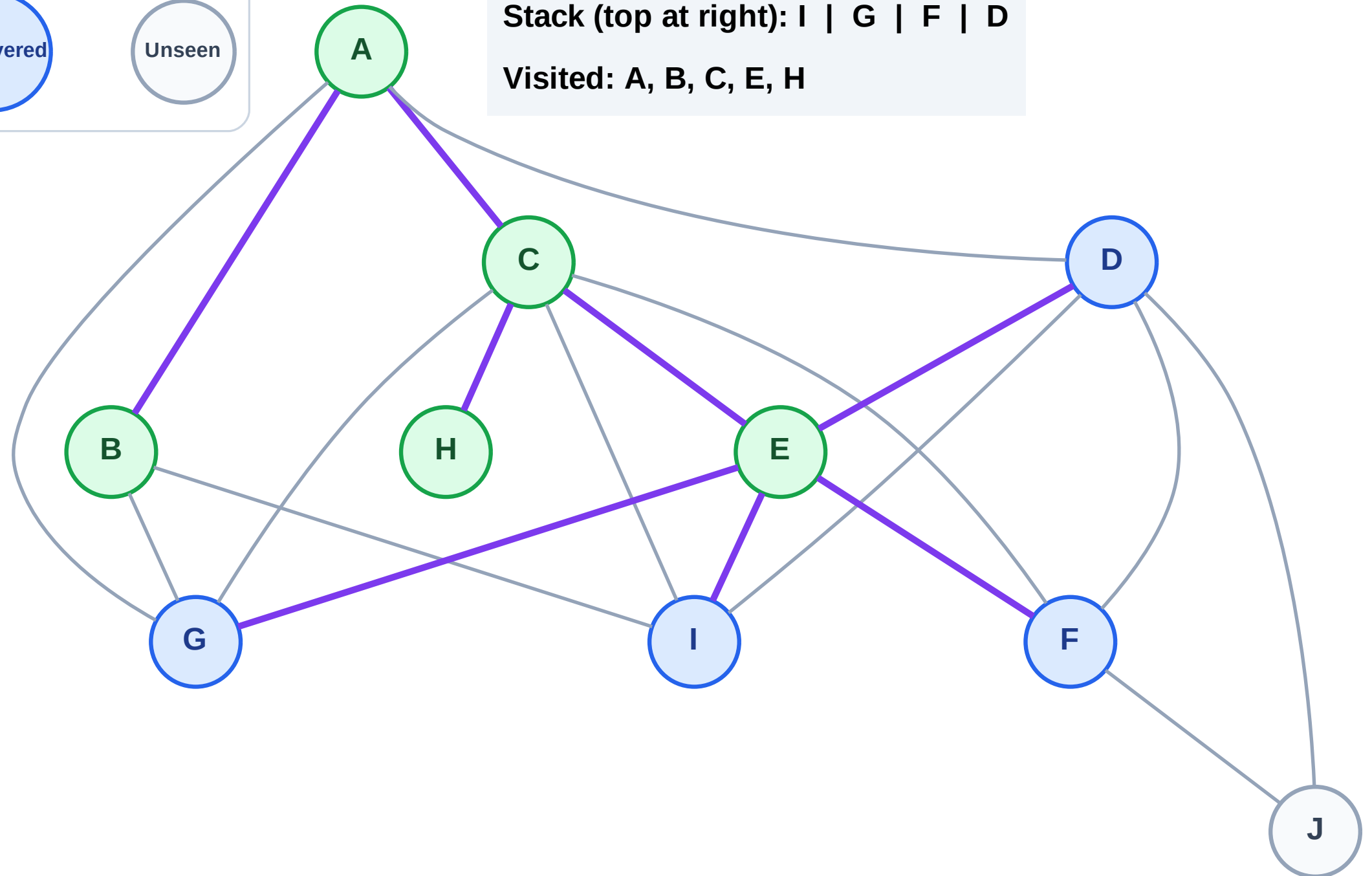
Step 11: No new neighbors from H

Legend



Stack (top at right): I | G | F | D

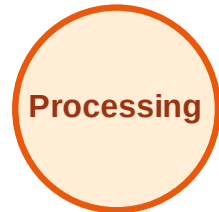
Visited: A, B, C, E, H



DFS Traversal

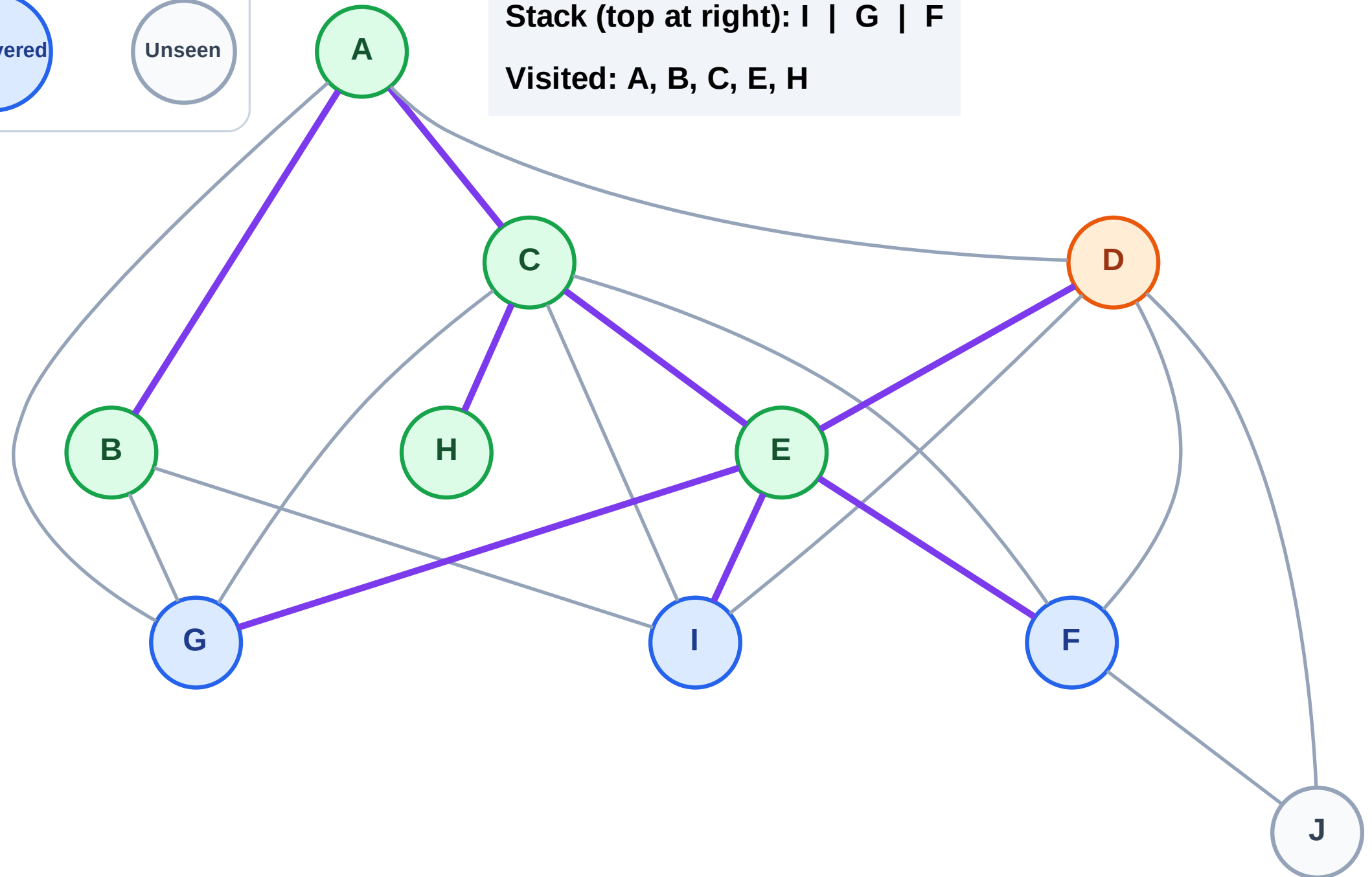
Step 12: Process node D

Legend



Stack (top at right): I | G | F

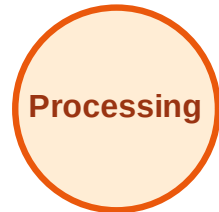
Visited: A, B, C, E, H



DFS Traversal

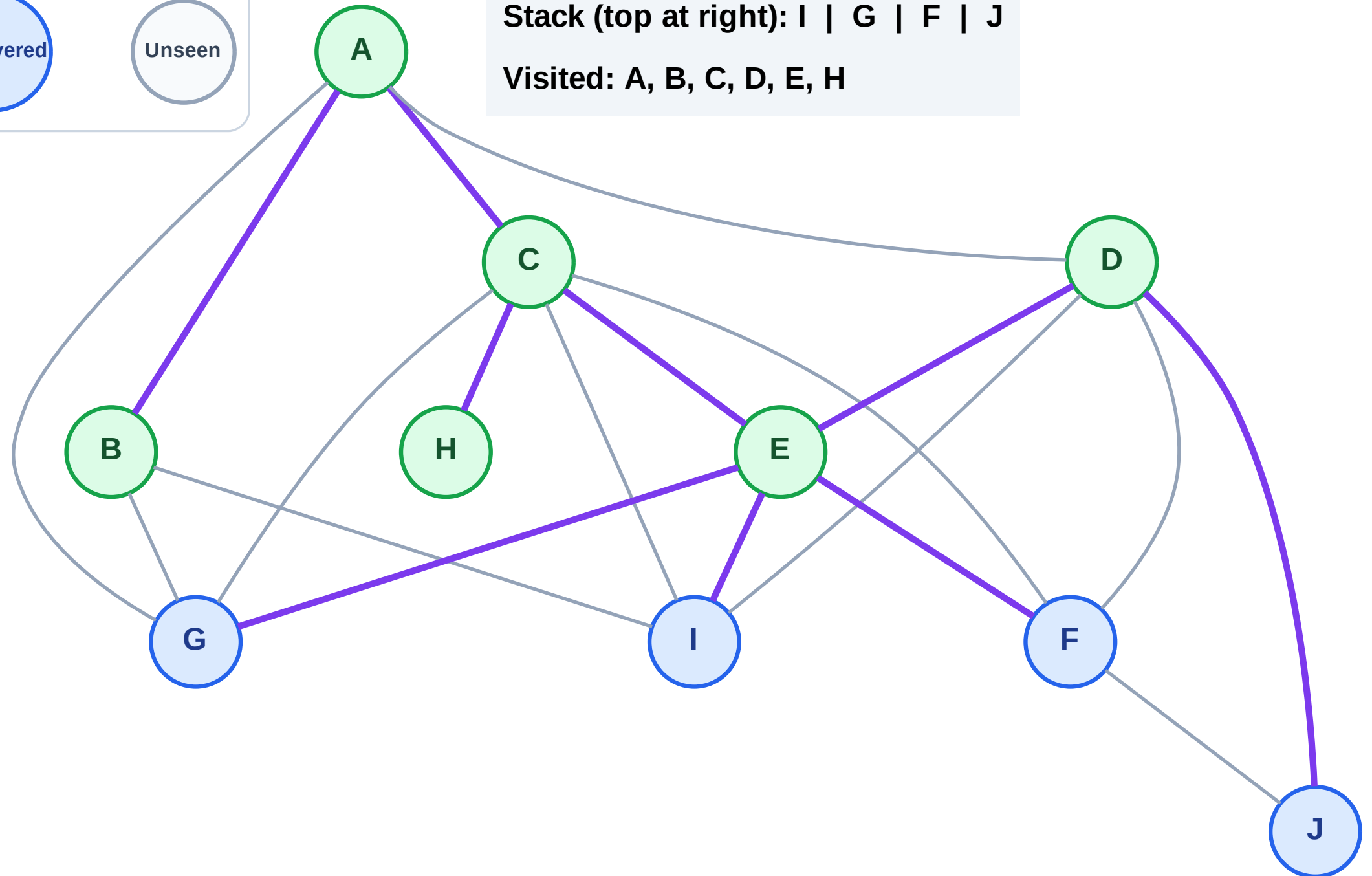
Step 13: Discover J from D

Legend



Stack (top at right): I | G | F | J

Visited: A, B, C, D, E, H



DFS Traversal

Step 14: Process node J

Legend

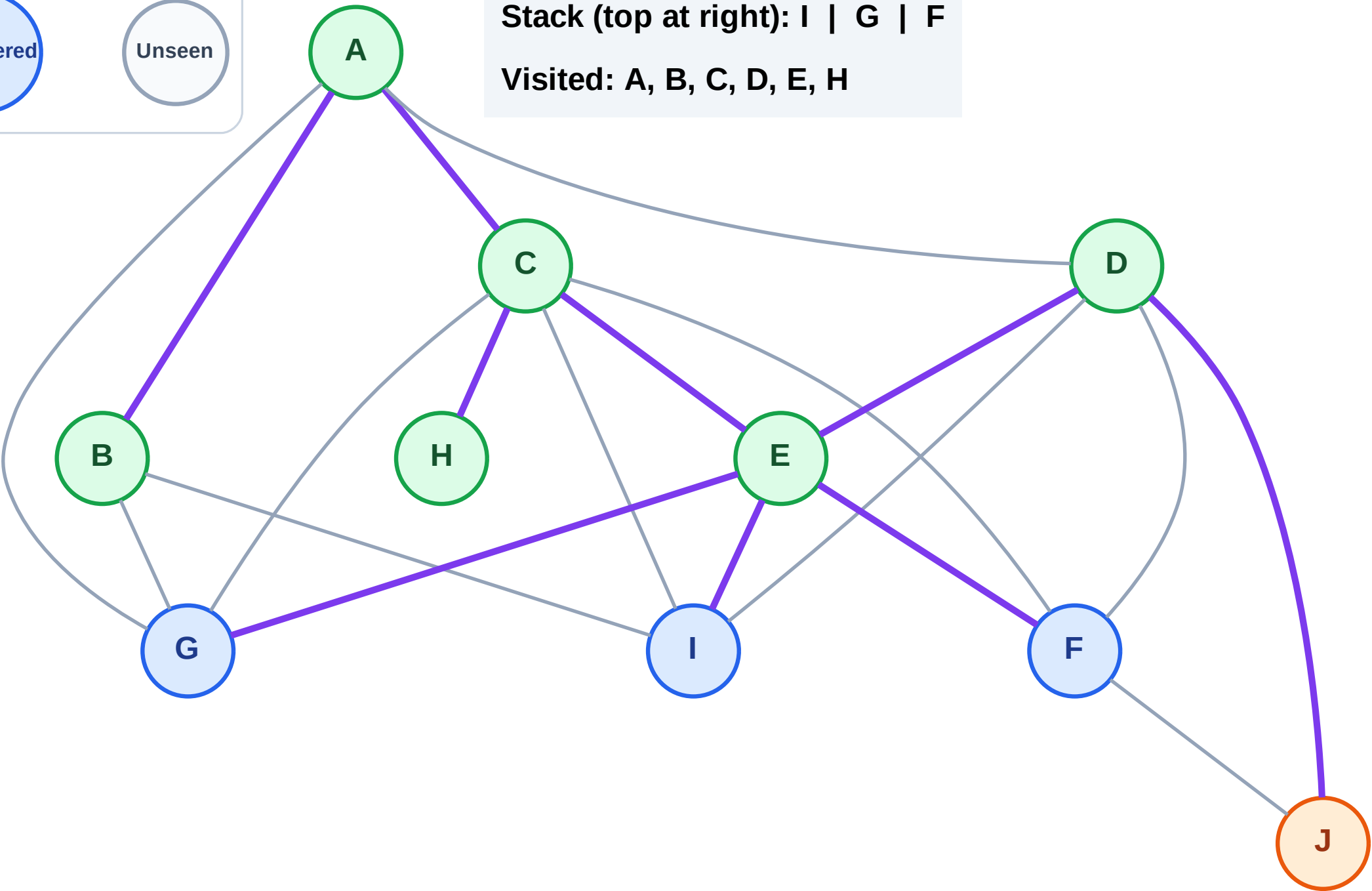
Complete

Processing

Discovered

Unseen

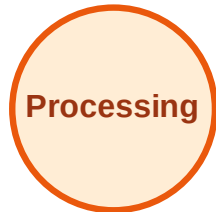
Stack (top at right): I | G | F
Visited: A, B, C, D, E, H



DFS Traversal

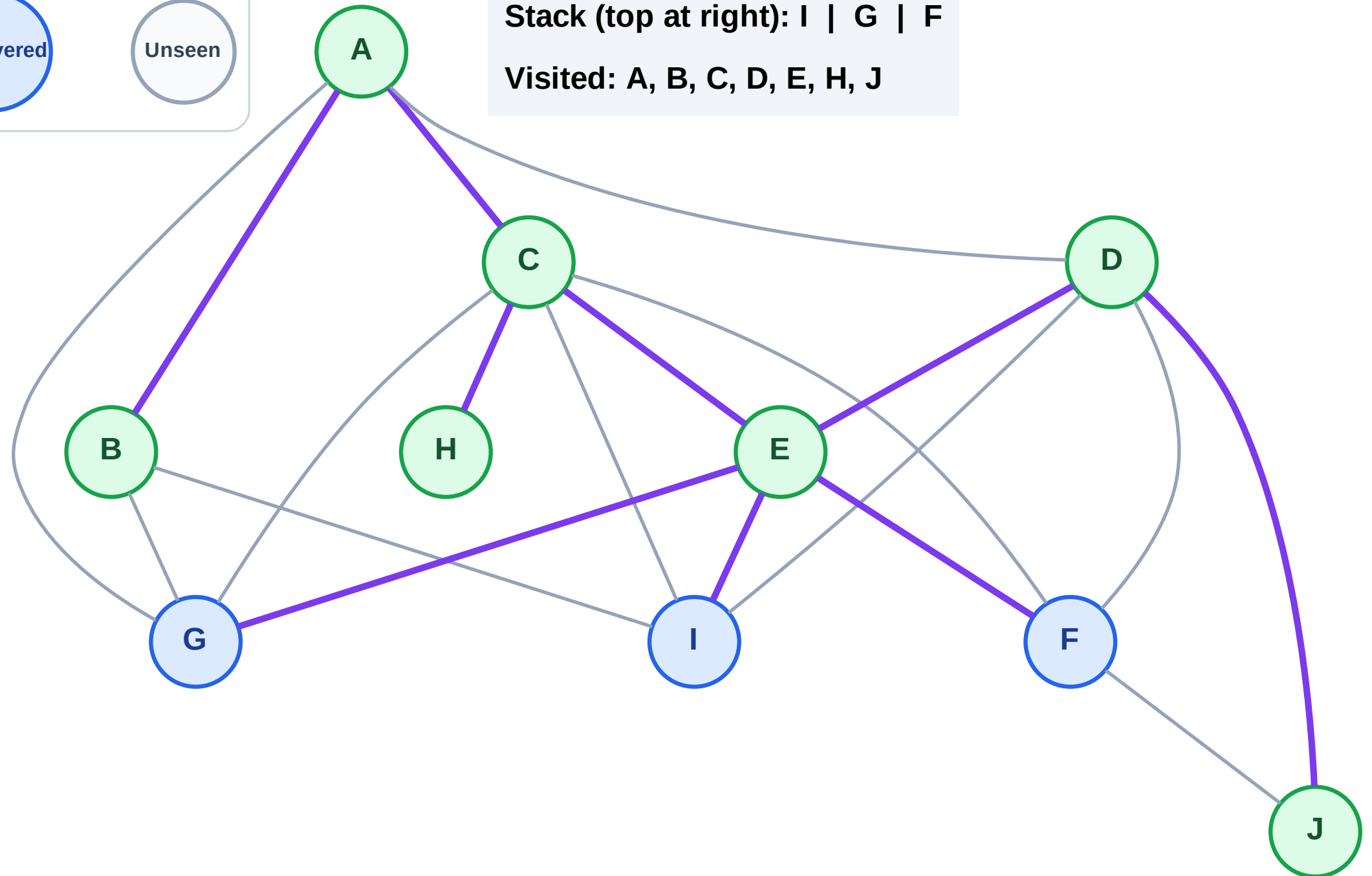
Step 15: No new neighbors from J

Legend



Stack (top at right): I | G | F

Visited: A, B, C, D, E, H, J



DFS Traversal

Step 16: Process node F

Legend

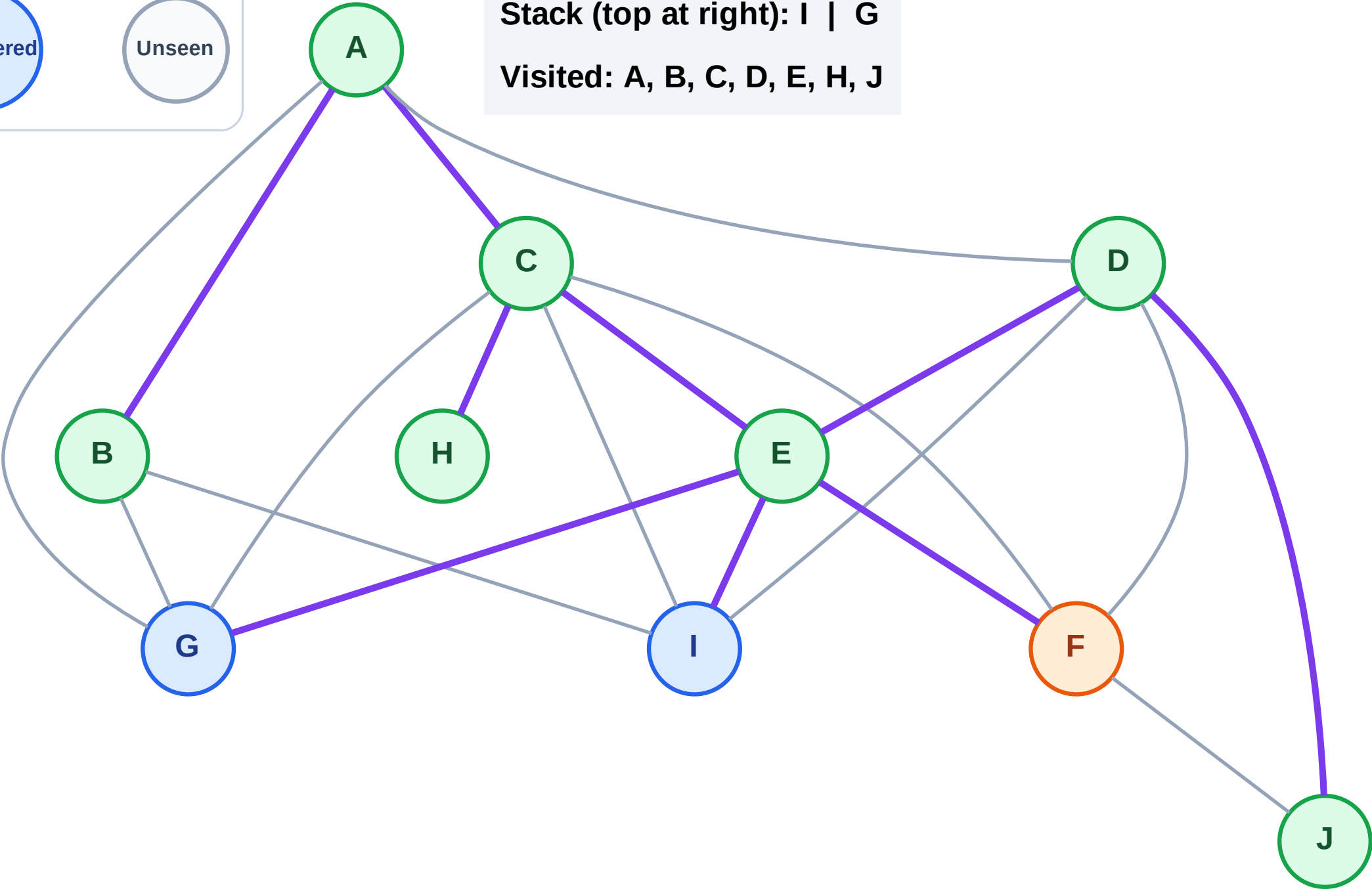
Complete

Processing

Discovered

Unseen

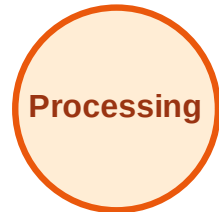
Stack (top at right): I | G
Visited: A, B, C, D, E, H, J



DFS Traversal

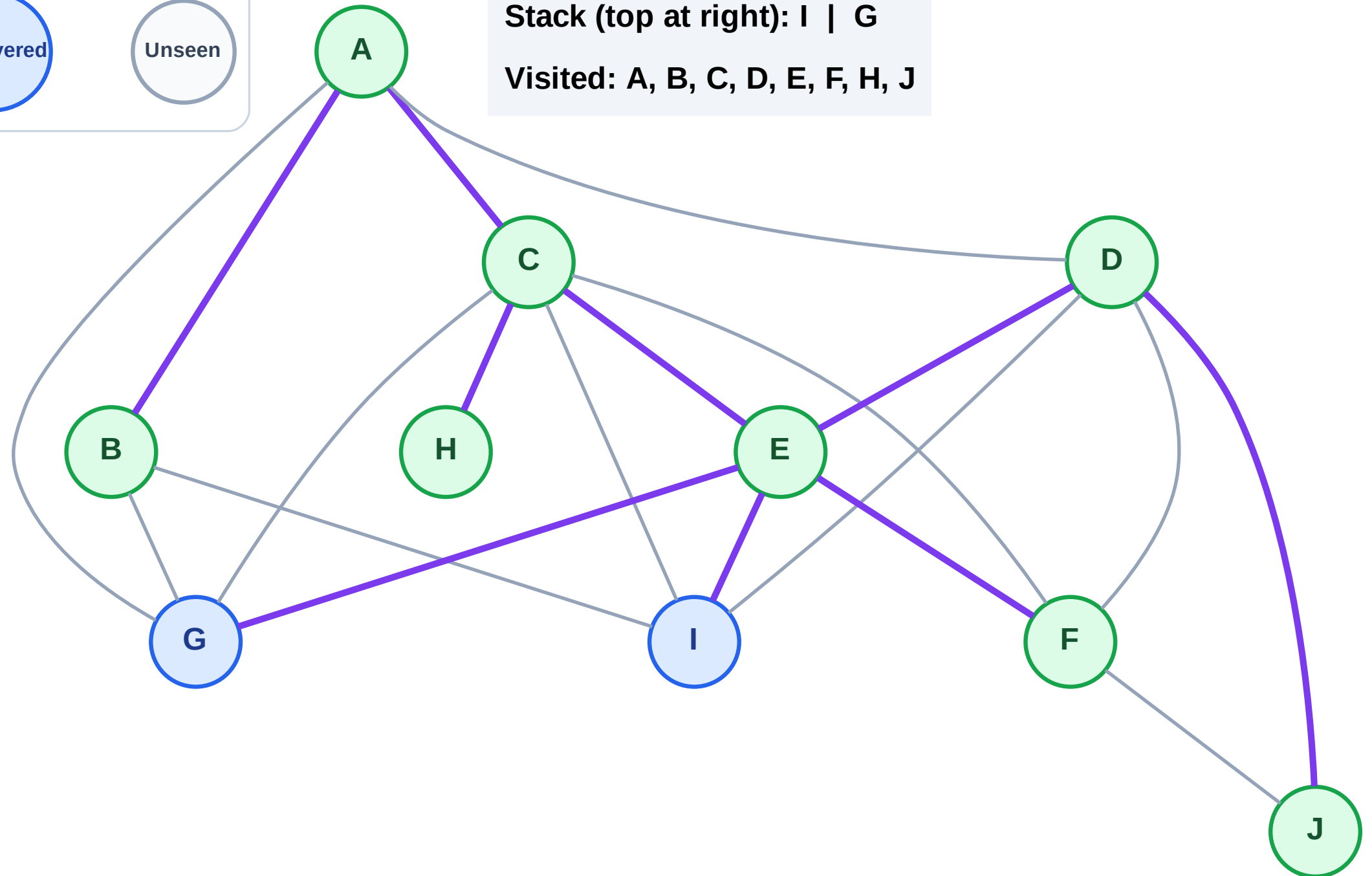
Step 17: No new neighbors from F

Legend



Stack (top at right): I | G

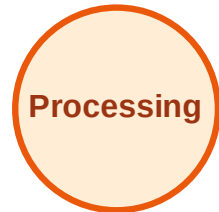
Visited: A, B, C, D, E, F, H, J



DFS Traversal

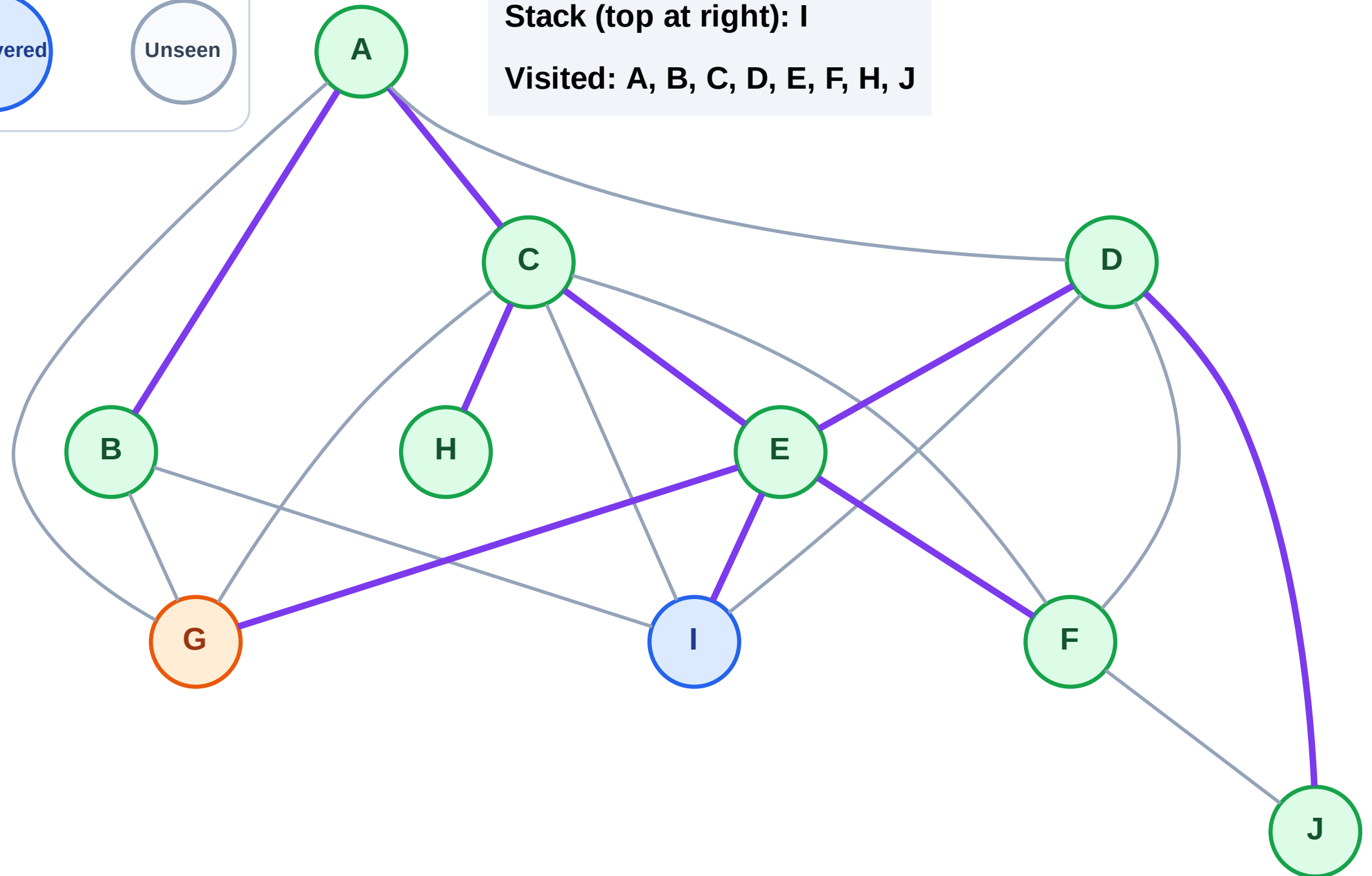
Step 18: Process node G

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, H, J



DFS Traversal

Step 19: No new neighbors from G

Legend

Complete

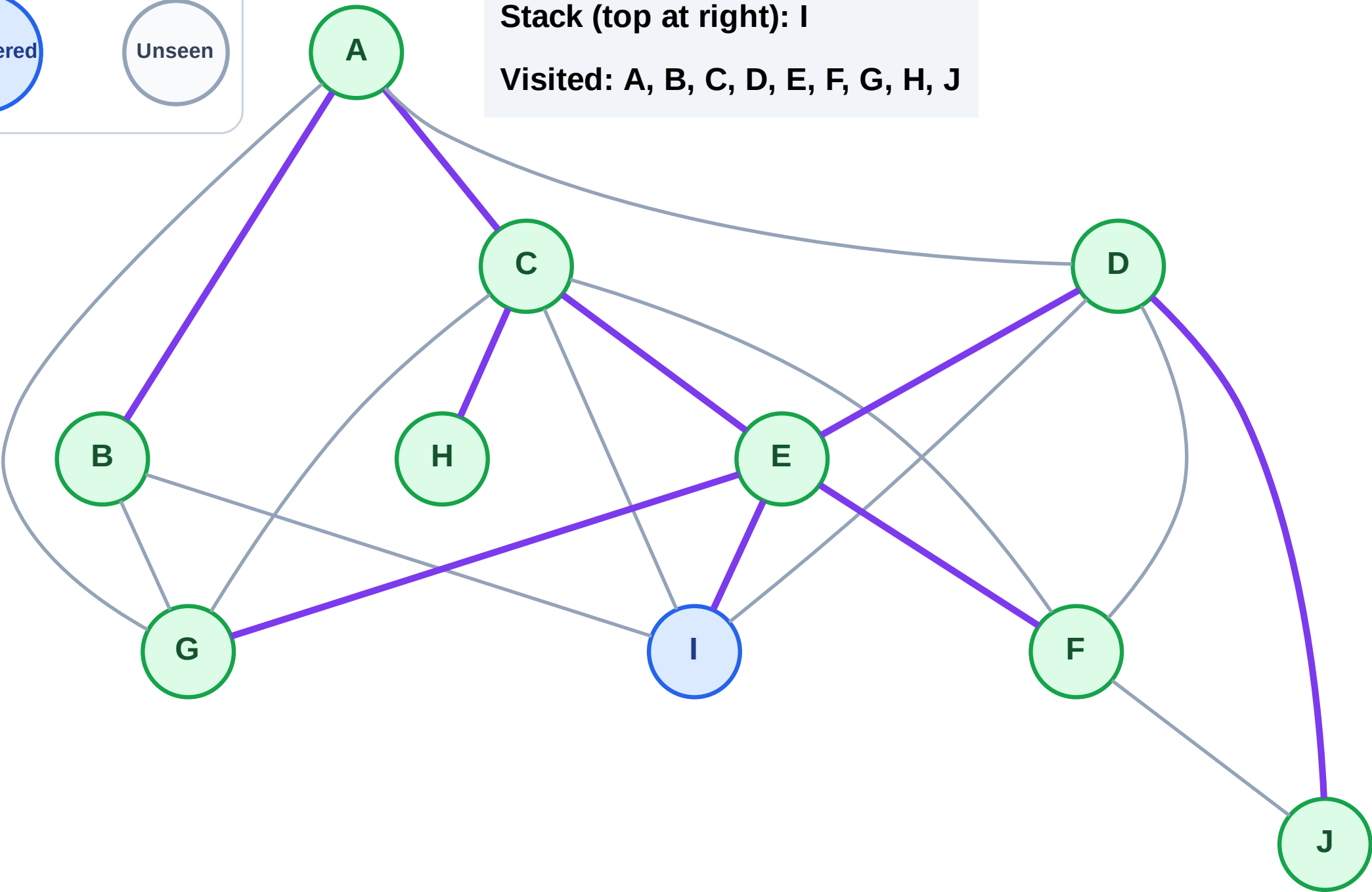
Processing

Discovered

Unseen

Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 20: Process node I

Legend

Complete

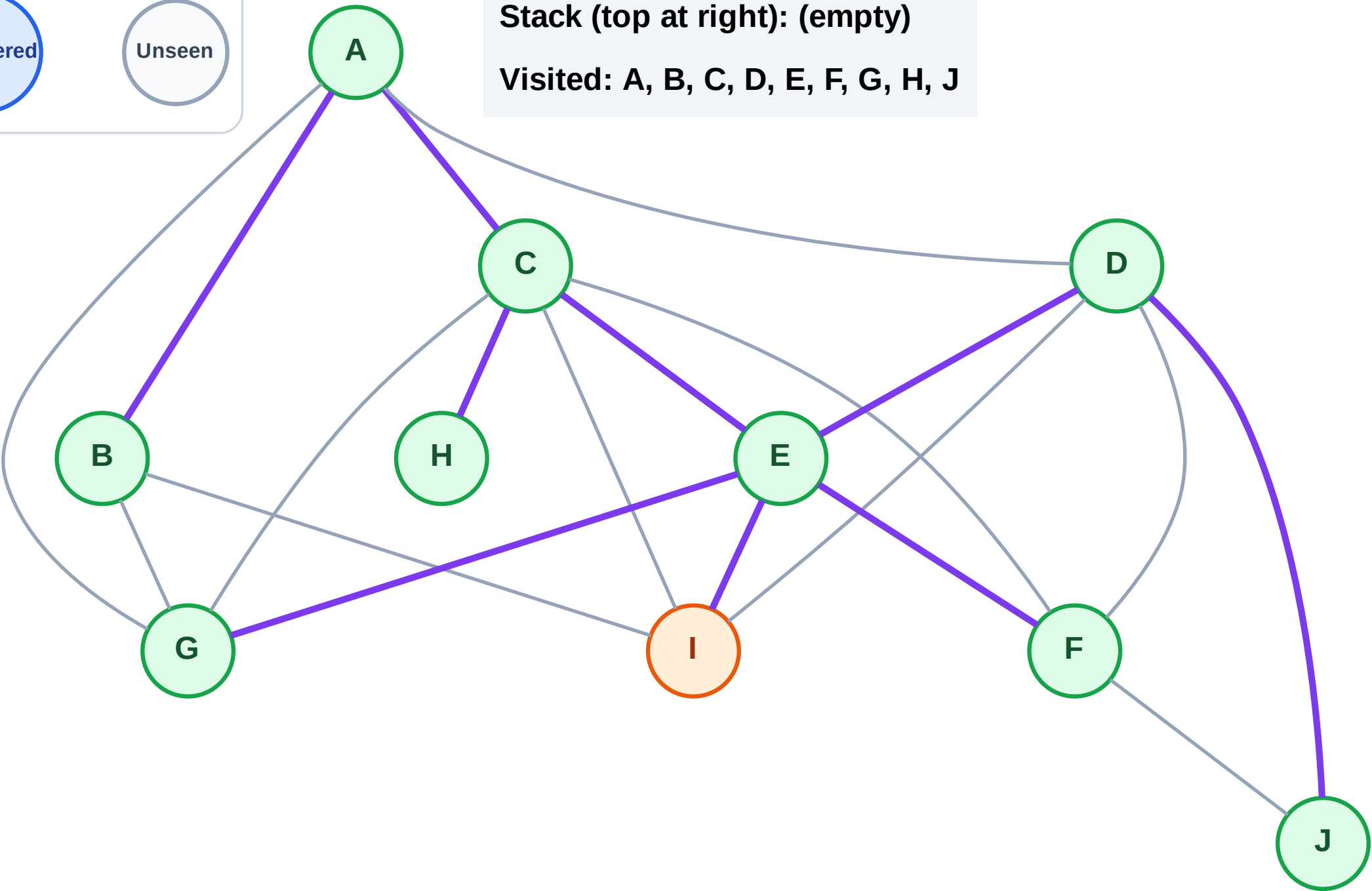
Processing

Discovered

Unseen

Stack (top at right): (empty)

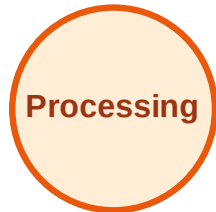
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

